Dolby Media Encoder

User's manual

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Contents

1 Introduction

This documentation describes how to use the Dolby Media Encoder , which is one of the software tools within the Dolby Media Producer Suite.

- About this documentation
- Abbreviated speaker notations
- Contacting Dolby

1.1 About this documentation

This documentation is intended for content professionals who want to use Dolby Media Encoder to encode a variety of source files (including Dolby Atmos master files) into a Dolby audio format, in preparation for delivery on various media targets.

Dolby Media Encoder supports encoding workflows targeted for several media targets:

- DVD-Video
- DVD-Audio
- Online Media
- Blu-ray Disc and Ultra HD Blu-ray

The following encoding formats are supported for .wav audio source files:

- Channel-based Dolby Digital
- Channel-based MLP Lossless
- Channel-based Dolby Digital Plus
- Channel-based Dolby TrueHD

The following encoding formats are supported for source Dolby Atmos master file sets:

- Dolby Digital Plus with Dolby Atmos and channel-based Dolby Digital Plus
- Dolby TrueHD with Dolby Atmos and channel-based Dolby TrueHD
- Channel-based Dolby Digital

Dolby Media Encoder presents a server-client application architecture.

- The Dolby Media Encoder Client application provides a user interface to configure and dispatch jobs to the Dolby Media Encoder Server.
- The Dolby Media Encoder Server receives and processes encoding jobs originated from Dolby Media Encoder Client instances.

By default, the client and server applications run locally within the same machine. The Dolby Media Encoder Client can also be configured to connect with a remote Dolby Media Encoder Server.

1.2 Abbreviated speaker notations

Dolby documentation uses specific names and abbreviations to describe speakers or speaker channels. Dolby notation can differ from the names and abbreviations used by other organizations and companies.

Note: This information is provided for reference only. Not all speaker outputs identified in the table are supported by Dolby Media Producer Suite.

Table 1: Dolby speaker notation correspondences

Dolby notation		CEA-861	ITU-R BS.2159-5 (Type B 10.2 channel)	SMPTE 2036-2 (22.2 channel)	IEC 62574 (30.2 channel)
L	Left	FL	L	FL	FL
R	Right	FR	R	FR	FR
C	Center	FC	C	FC	FC
Ls	Left surround	RL	LS	—	LS, LSd
Rs	Right surround	RR	RS	—	RS, RSd
Lrs	Left rear surround Left back	RLC	LB	BL	BL
Rrs	Right rear surround Right back	RRC	RB	BR	BR
Cs	Center surround	RC	—	BC	BC
Lw	Left wide	FLW	—	—	FLW
Rw	Right wide	FRW	—	—	FRW
Lc	Left center	FLC	—	FLC	FLC
Rc	Right center	FRC	—	FRC	FRC
Ls1	Left surround 1	—	—	SiL	SiL
Rs1	Right surround 1	—	—	SiR	SiR
Lfh Lvh	Left front height	FLH	LH	TpFL	TpFL
Rfh Rvh	Right front height	FRH	RH	TpFR	TpFR
Ltm Lts	Left top middle Left top surround	—	—	TpSiL	TpSiL
Rtm Rts	Right top middle Right top surround	—	—	TpSiR	TpSiR
Ltf	Left top front	—	—	—	—
Rtf	Right top front	—	—	—	—
Ltr	Left top rear	—	—	—	TpLS
Rtr	Right top rear	—	—	—	TpRS
Lrh	Left rear height	—	—	TpBL	TpBL
Rrh	Right rear height	—	—	TpBR	TpBR
LFE LFE1	Low-frequency effects 1	LFE	LFE1	LFE1	LFE1
LFE2	Low-frequency effects 2	—	LFE2	LFE2	LFE2
Lsc	Left screen	—	—	—	—
Rsc	Right screen	—	—	—	—
Ls2	Left surround 2	—	—	—	—
Rs2	Right surround 2	—	—	—	—

Table 1: Dolby speaker notation correspondences (continued)

Dolby notation		CEA-861	ITU-R BS.2159-5 (Type B 10.2 channel)	SMPTE 2036-2 (22.2 channel)	IEC 62574 (30.2 channel)
Lcs	Left center surround	—	—	—	—
Rcs	Right center surround	—	—	—	—
Ch	Center height	FCH	—	TpFC	TpFC
Ts	Top surround	TC	—	TpC	TpC
Lbin	Left binaural	—	—	—	—
Rbin	Right binaural	—	—	—	—
Lo	Left only	—	—	—	—
Ro	Right only	—	—	—	—
Lsd	Left surround direct	—	—	—	—
Rsd	Right surround direct	—	—	—	—
Le	Left Dolby Atmos enabled	—	—	—	—
Re	Right Dolby Atmos enabled	—	—	—	—
Lse	Left surround Dolby Atmos enabled	—	—	—	—
Rse	Right surround Dolby Atmos enabled	—	—	—	—
Lrse	Left rear surround Dolby Atmos enabled	—	—	—	—
Rrse	Right rear surround Dolby Atmos enabled	—	—	—	—
—		—	—	BtFC	BtFC
—		—	—	BtFL	BtFL
—		—	—	BtFR	BtFR
—		—	CH	TpBC	TpBC

1.3 Contacting Dolby

You can contact Dolby regarding this product and its supporting documentation.

For technical or product-related questions, contact DMPsupport@dolby.com.

For comments about this documentation, contact documentation@dolby.com.

2 Getting started with Dolby Media Encoder

The Dolby Media Encoder Client and Dolby Media Encoder Server applications are part of the Dolby Media Producer Suite. Both applications are installed when the full suite is installed. Refer to the *Dolby Media Producer Suite release notes* for system requirements and installation steps.

- Launching Dolby Media Encoder
- Quitting Dolby Media Encoder
- Checking the software version
- Application user interface

2.1 Launching Dolby Media Encoder

The Dolby Media Encoder Client application provides access to all Dolby Media Encoder user-facing controls.

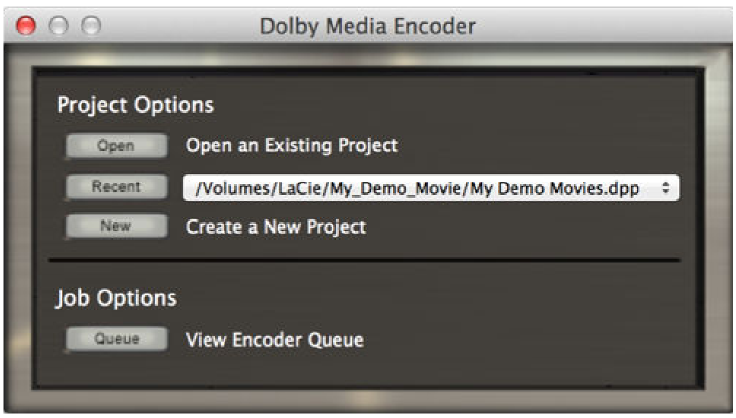
Procedure

To launch the application, do one of the following actions:

- Double-click the Dolby Media Encoder Client application file (located by default in /Applications/Dolby/Dolby Media Producer/).
- Click the Dolby Media Encoder Client application icon.

Results

The Dolby Media Encoder initial window appears.



2.2 Quitting Dolby Media Encoder

You can quit the Dolby Media Encoder session by using the **Dolby Media Encoder** menu or icon.

Procedure

To quit the application, do one of the following actions:

- In the **Dolby Media Encoder** menu bar, select **Dolby Encoder Client** > **Quit Dolby Media Encoder** .
- Right-click the **Dolby Media Encoder** icon, and select **Quit**.

2.3 Checking the software version

You can check the version number of the Dolby Media Encoder, including the version number of the parent Dolby Media Producer Suite.

Procedure

Select **Dolby Media Encoder** > **About Dolby Media Encoder** .

Note: You can also access the version number by clicking the Dolby logo at the upper-left corner of a job editor window.

2.4 Application user interface

The Dolby Media Encoder Client makes consistent use of a series of conventions in its user interface.

The Dolby Media Encoder runs on a series of independent windows, some of which include a series of tabs containing additional options and settings.

Actions are triggered by buttons, drop-down menu selections, and status switches. Each of these UI elements can be displayed as inactive, active or unavailable.

Table 2: UI conventions for buttons and status switches

Button	Status switch	Status
		Inactive
		Active
		Unavailable

Some of the Dolby Media Encoder windows and tabs may display entire unresponsive sections that appear grayed out to the user. Such areas indicate settings and options that cannot be edited under the current encoding configuration.

3 Remote Dolby Media Encoder Server setup

You can configure the Dolby Media Encoder Client to send jobs to remote servers. This requires mounting and configuring file paths accordingly.

- Adding and removing remote Dolby Media Encoder servers
- Checking Dolby Media Encoder Server versions
- Setting up file paths for remote servers
- Troubleshooting server connections

3.1 Adding and removing remote Dolby Media Encoder servers

You must add remote Dolby Media Encoder servers to your local Dolby Media Encoder Client before you can access and send encoding jobs to a remote server.

Prerequisites

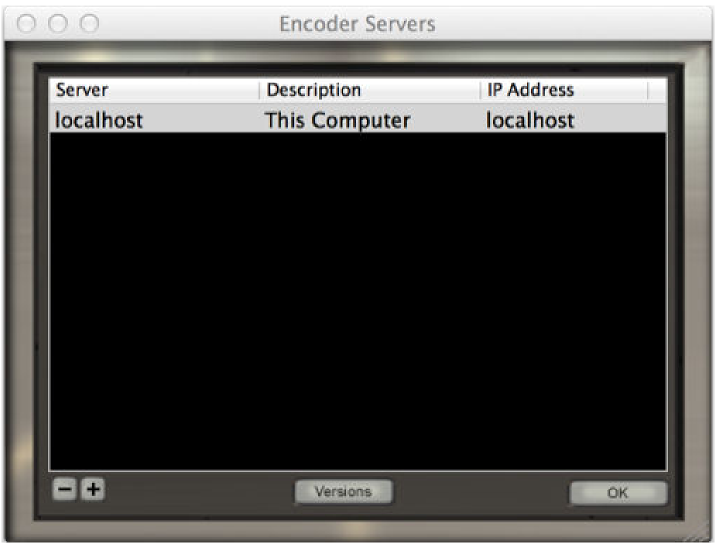
- Make sure that both client and server have their host names properly configured.
- Make sure that you know the IP address of your server. You can find it under **System Preferences** > **Network** on the server machine.
- Make sure that you know the network name of your server. You can find it under **System Preferences** > **Sharing** on the server machine.

Procedure

1. On the Dolby Media Encoder Client menu bar, select **Settings** > **Encoder Servers**.

   The application shows a window listing all servers currently added to your network.

   Figure 1: Encoder Servers window

2. In the **Encoder Servers** window, add a new server entry by clicking the  button at the lower left of the window.

The application adds a new row to the server list.

3. Enter the IP address or network name of the server (the machine hosting the Dolby Media Encoder Server application) by doing one of the following actions:
   - On the new row of the server list, double-click **new server**, and enter the IP address or network name.
   - Right-click the new row of the server list, and select from the list of available IP addresses and host names.
4. Optional: If you want to change a server description, double-click the default description (**remote encoder server**) and type your description.
5. To remove an existing server, do the following actions:
   a) Select the server you wish to remove by clicking on its entry.
   b) Click the  button at the lower left of the window.

What to do next
When you have finished adding or removing remote servers, click **OK** to close the **Encoder Servers** window.

3.2 Checking Dolby Media Encoder Server versions

You can check the version number of any Dolby Media Encoder Server added to your client application.

Procedure

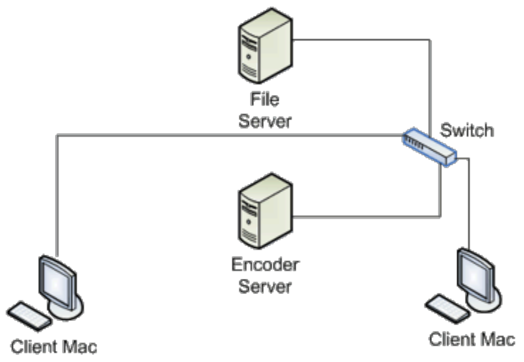
1. On the Dolby Media Encoder Client menu bar, select **Settings** > **Encoder Servers**.
2. From the server list, select a server by clicking its entry.
3. At the bottom of the **Encoder Servers** window, click **Versions**.
   The application shows a new window displaying the version number of the server.

3.3 Setting up file paths for remote servers

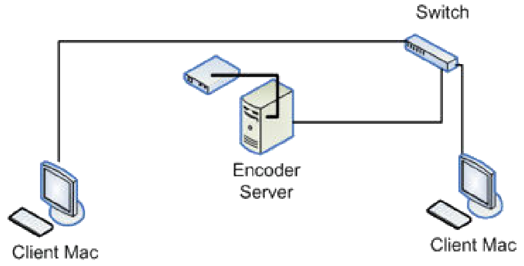
When encoding through remote Dolby Media Encoder servers, both client and server must know the location path for source and destination files.

About this task
Source and destination files for Dolby Media Encoder can be stored across various locations. These locations include:

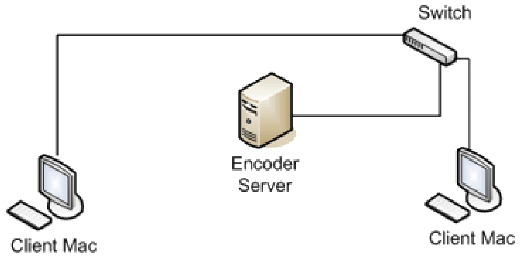
- Remote file servers, separate from both encoder server and encoder client machines



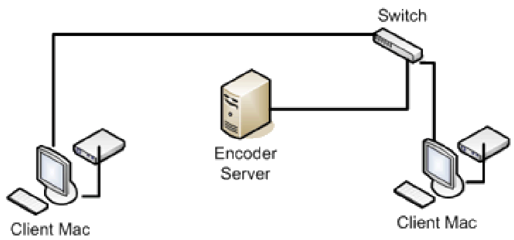
- External drives attached to a remote encoder server



- Internal drives within a remote encoder server



- External drives attached to an encoder client



- To set up file paths for a remote file server:

  a) Manually mount the file server on both client and server machines.

  b) Verify that file path automounting is enabled.

  Client and server share the same file paths.

- To set up file paths for external drives attached to a remote encoder server:

  a) Ensure that the server external drive is designated as a shared drive.

  b) Manually mount the server external drive on the client.

c) Verify that file path automounting is enabled.

Client machines access the files through the server.

- To set up file paths for internal drives within a remote encoder server:

  a) Manually mount the server internal drive on the client.

  b) Manually map the file paths.

  Mapping must be performed manually because server and client paths are not the same. For example:

  - Local path: /Volumes/X-Serve/waves
  - Server name: 10.111.111.111
  - Server path: /Volumes/Files/G-Raid/waves

  In this example, G-Raid is the name of the internal drive.

- To set up file paths for external drives attached to an encoder client:

  a) Ensure that the client external drive is designated as a shared drive.

  b) Manually mount the client external drive on the server.

  This step must be performed on the server machine.

  c) Verify that file path automounting is enabled.

  d) If automounting does not work, manually map file paths so that the server can find the client drive.

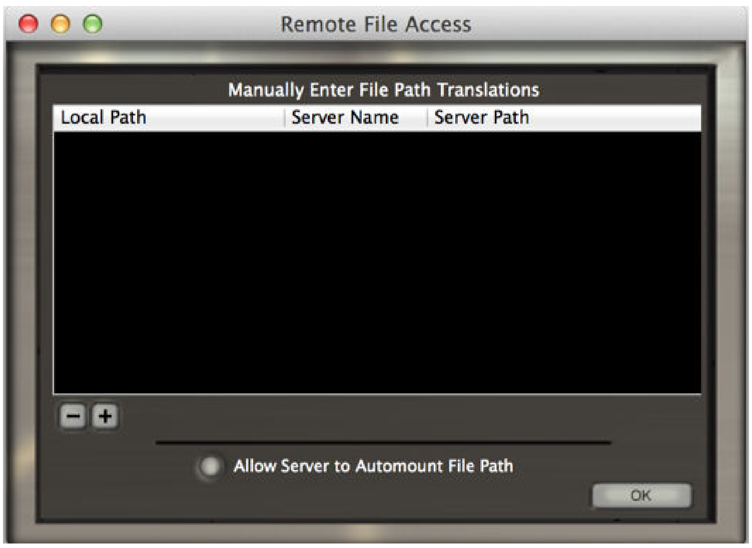
3.3.1 Configuring file path automounting

We recommend that you enable the automatic detection and mounting of source and destination file paths.

Procedure

1. On the Dolby Media Encoder Client menu bar, select **Settings** > **Remote File Access**.

   The system opens the **Remote File Access** window.



2. To activate or deactivate automounting, click **Allow Server to Automount File Path**.
3. To close the window, click **OK**.

What to do next

If your source or destination files are located on a remote machine, you must manually mount those remote locations.

Related information

Mounting remote locations manually on page 15

3.3.2 Mounting remote locations manually

You can manually mount remote source and destination locations so that they are accessible on your client and server machines.

Prerequisites
Remote locations need to be configured to be visible in the network. Refer to your operating system documentation for further information.

Procedure

1. On the menu bar of the Finder application, select **Go** > **Connect to Server**.
2. Select to the desired remote machines.
   - In the **Connect to Server** window, click **Browse**.
   - In the **Server Address** text field at the top of the window, manually type the directory path.
3. To finish, click **Connect**.

Related information

Configuring file path automounting on page 14

3.3.3 Mapping file paths manually

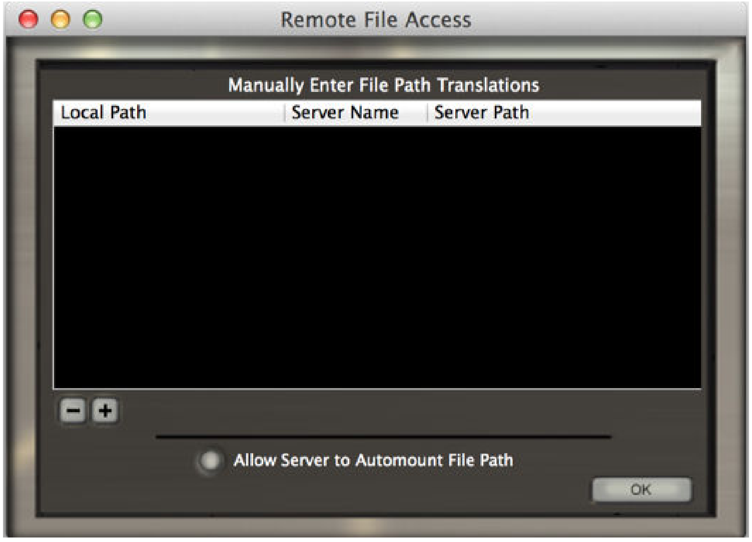
You can manually enter file path information in the Dolby Media Encoder Client **Remote File Access** window.

About this task
Each mapped path in Dolby Media Encoder is defined by a file path translation entry. You can create multiple path entries for each remote server.

Procedure

1. On the Dolby Media Encoder Client menu bar, select **Settings** > **Remote File Access**.

   The system opens the **Remote File Access** window.



2. Click the  button.

   The system displays the **Path Translation Entry** window.

3. In the **Client Path** field, enter the file path of the client side.

   Clicking the  (browse) button to the right of the **Client Path** field opens the Finder on the client machine so you can easily select a folder.
4. From the list on the **Server** drop-down menu, select the server that will use the path translation entry.
5. In the **Server Path** field, enter the file path for the server side.

   This path should correspond to the same location provided in the **Client Path**, but seen from the server side. You need to browse the location on the server machine to identify the exact server side file path.
6. To save the entry, click **OK**.

3.4 Troubleshooting server connections

Dolby Media Encoder Client may present a `Cannot connect to server` error. You can solve this issue in several ways.

- Verify the server name.

If the client and the server are running on the same machine, the server name must end in .local.

- Verify that an iLok USB key is inserted into the machine running Dolby Media Encoder Server.
- Make adjustments in the **Encoder Servers** window, which is accessible through the **Settings** menu.

  Right-clicking an entry on the list should present you a drop-down menu with server options. Re-selecting a server from this list updates its IP address information, which may have changed if the machines are connected to a dynamic IP network.
- Verify that your network is properly configured.

  Failing to see any servers when right-clicking an entry in the **Encoder Servers** window indicates a network issue.
- Reboot the client machine.

  Client machines can lose connectivity with a remote server after a server is rebooted. Rebooting the client machine forces it to reconnect with the server.
- Reboot the server and client machines.

  The Dolby Media Encoder Server application is designed to restart automatically if it crashes. This automatic restart may fail if the crash takes place during an encoding job.

4 Dolby bitstream encoding with Dolby Media Encoder

Dolby Media Encoder can encode audio source files or Dolby Atmos master file sets to supported Dolby audio formats, in preparation for delivery on various media targets.

4.1 Encoding workflow in Dolby Media Encoder

Dolby recommends following a linear workflow when setting up individual encoding jobs in the Dolby Media Encoder Client interface.

About this task

You can perform all of the following actions from the **job editor** window.

Procedure

1. Create a new project, or add a new job to an existing project.
2. Select an encoding format and media target.
3. Select a particular encoding configuration.
4. Load the source files required for the selected encoding configuration.
5. Further configure the encoder settings, if needed.
6. Enter user information for the encoding job, if needed.
7. Submit the encoding job to Dolby Media Encoder Server.

Related information

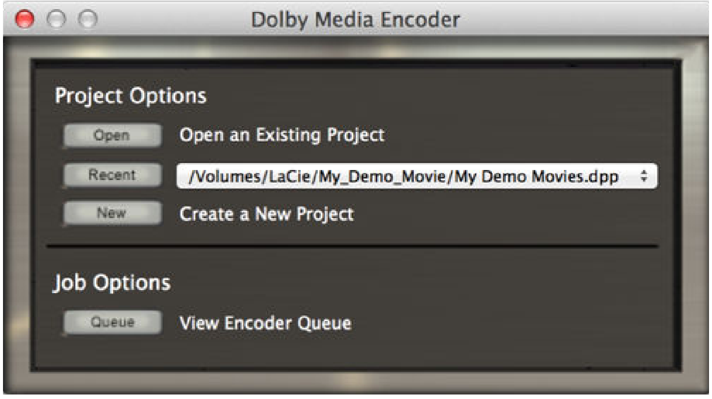
Encoding files through Hot Job folders on page 78

4.1.1 Creating new projects and jobs

Dolby Media Encoder groups job records into projects. Projects store information for any encoding job created, including jobs that have not been submitted for encoding.

Procedure

1. Launch the Dolby Media Encoder Client application.
   The system displays the Dolby Media Encoder main window.



2. In the **Project Options** section, click **New**.
   The system displays the **project properties** window.

3. In the **Project Name** field, enter a name for the project.

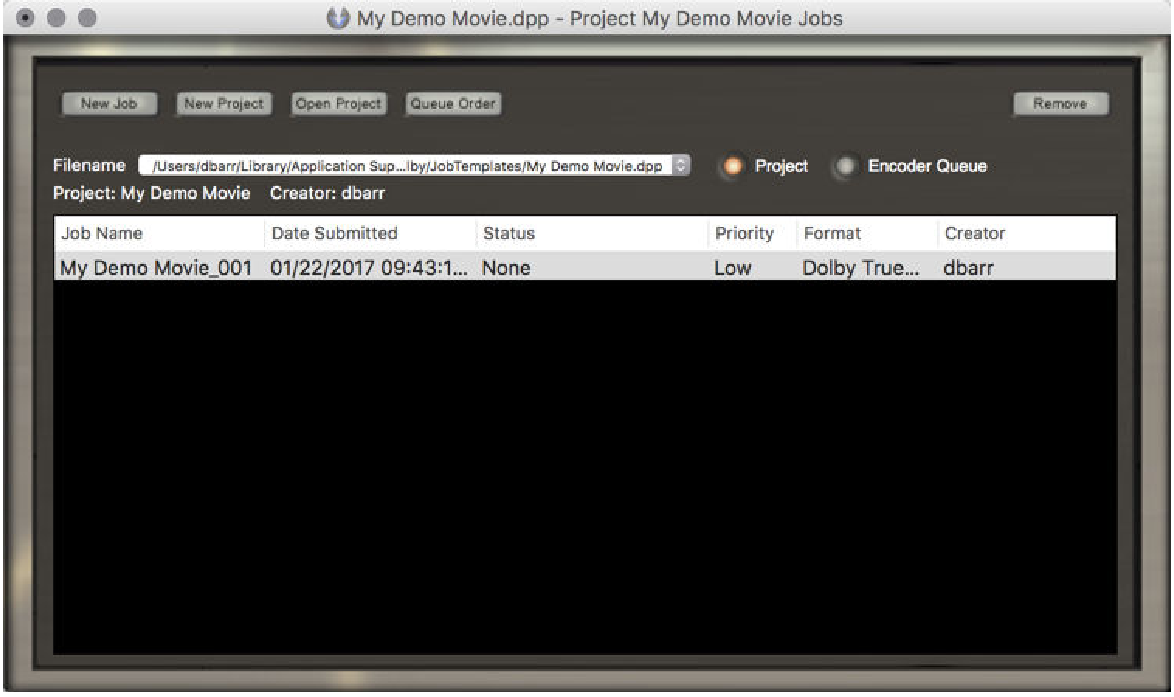
   The remaining fields in the **project properties** window are optional.
4. Click **Create**.
   The system displays a **Save** dialog box.
5. Save the project .dpp file at a location of your choice.

   The file name can be different from the project name.
6. Click **OK**.

Results

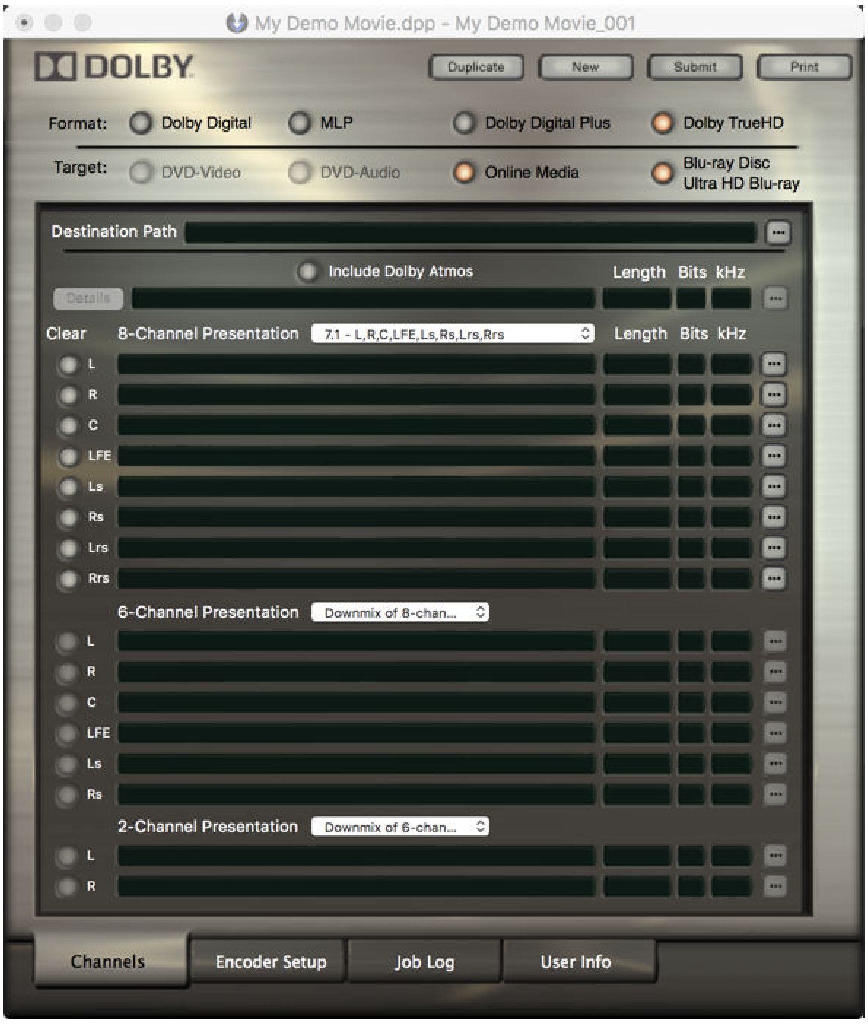
The system opens the **job list** window and a **job editor**.

Figure 2: Job list window



Note: The **job list** window shows the project name on the title bar.

Figure 3: Job editor



Note: The **job editor** window shows the job name on the title bar.

The **job list** includes an initial encoding job, named <project_name>_001. You can start configuring this job within its **job editor** window.

4.1.2 Choosing an encoding format and media target

Encoding jobs are created for specific Dolby audio formats and media targets. The encoding options available within the **job editor** window vary depending on the selected format and media target.

Procedure

1. In the **Format** section of the **job editor** window, select a supported Dolby audio format.
   - Dolby TrueHD (default)
   - Dolby Digital Plus
   - MLP
   - Dolby Digital
2. In the **Target** section of the **job editor** window, select media targets.

   The available media targets are determined by your selected encoding format.

   a) For **Dolby TrueHD** encoding, **Blu-ray Disc and Ultra HD Blu-ray** and **Online Media** media targets are both always selected.

   You can also use Dolby TrueHD streams formatted for Blu-ray Disc in Online Media.

   b) For **Dolby Digital Plus** encoding, you can select **Blu-ray Disc and Ultra HD Blu-ray** or **Online Media** media targets.

   You can select only one media target.

   c) For **MPL** encoding, you can only select **DVD-Audio** as media target.

   d) For **Dolby Digital** encoding, you can select **DVD-Video** and **Blu-ray Disc and Ultra HD Blu-ray** media targets.

   You can select either target individually, or both.

   When two media targets are selected, the system produces a single encoded file compatible with both targets. The attributes of multiple-target files are restricted to those common to both targets.

4.1.3 Selecting encoding configuration and loading source files

The types of source material that can be used to create an encoded file vary according to the selected encoding configuration.

About this task

The format and targets selected for an encoding job determine its available configuration options. Each encoding configuration requires a particular set of source materials.

Dolby Media Encoder supports source .wav, .aif, or .atmos master files, depending on the selected encoding configuration.

Note: Dolby Media Producer Suite v2.4 introduced support for Broadcast Wave Format files with Audio Definition Model metadata.

Procedure

1. Select an encoding configuration by modifying the options available in the **Channels** tab of the job editor window.
2. Select a source file for each available source field.
   a) Click the (browse) button to the right of the source field.
   b) In the browse window, locate and highlight the source file.
   c) In the browse window, click **OK**.

   Dolby Media Encoder auto-populates channel-based source fields when the source files follow a particular file-naming convention. Selecting one file in the set populates the remaining source fields.

   Once you have selected your source files, Dolby Media Encoder will keep them selected if you switch between encoding formats, performing any required adjustments in case the original channel layout is not supported in the new format. This applies to both channel-based files and Dolby Atmos master files.
3. To remove any of the selected source files, click **Clear** to the left of a particular source field.

   Holding the Ctrl key while clicking the **Clear** button clears all loaded source files.

Related information

Dolby Digital encoding configurations on page 65
Dolby Digital Plus encoding configurations on page 50
Dolby TrueHD encoding configurations on page 37
MLP encoding configurations on page 60
File-name conventions for automatic file selection on page 79

4.1.4 Supported source material

Dolby Media Encoder supports encoding of .wav or .aif PCM source files at several sample rates, depending on the selected encoding format.

Note: Dolby Media Encoder supports Dolby Atmos master files with 48 kHz sample rates only.

Dolby TrueHD
Supports channel-based encoding at 44.1, 48, 88.2, 96, 176.4, and 192 kHz.

Dolby Digital Plus
Encodes natively at 48 kHz. Content at 44.1, 88.2, 96, 176.4, and 192 kHz is automatically converted to 48 kHz.

MLP
Supports encoding at 44.1, 48, 88.2, 96, 176.4, and 192 kHz.

Dolby Digital
Encodes natively at 48kHz. Content at 44.1, 88.2, 96, 176.4, and 192 kHz is automatically converted to 48 kHz.

Note: The Blu-ray Disc specification does not support 44.1, 88.2, or 176.4 kHz.

4.1.5 Specifying name and destination path for encoded files

After an encoding job has its source files loaded, Dolby Media Encoder automatically fills in the encoded file destination path, name, and extension. You can change the default destination path and file name on the **Channels** tab of the job editor window.

About this task

- The destination path defaults to the source file directory.
- The encoded file name defaults to the name of the first source file listed.
- The encoded file extension is automatically set according to the selected format. The extension updates dynamically if the encoding format is changed.
- Click the  (browse) button to the right of the **destination path** field.

  a) Navigate the browse window and select a new location.

  b) Type in a new file name.

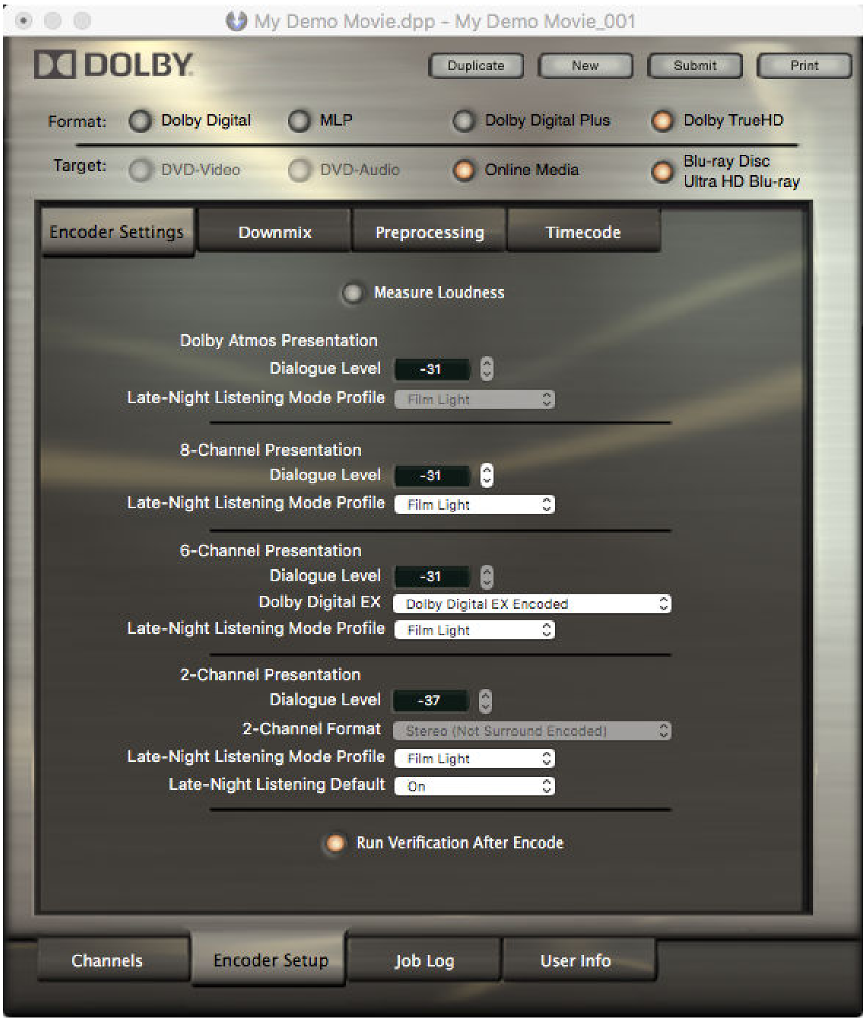
  c) Click **Save**.
- Alternatively, type or paste the destination path and file name in the **destination path** field.

4.1.6 Configuring additional encoder settings

Dolby Media Encoder automatically configures each encoder setting and metadata parameter required for the encoding process to industry-standard default values, based on the selected encoding format. You can make changes to these settings on the **Encoder Setup** tab of the job editor window.

About this task

Different configuration options are available in each of the tabs within the **Encoder Setup** tab of the **job editor** window.



Procedure

1. Configure the options in the **Encoder Settings** tab to adjust dialogue level, data rate, dynamic range control, and other parameters applicable to the selected encoding format.

   The **Encoder Settings** tab contains the most critical configuration options.

2. Adjust downmix settings on the **Downmix** tab.

   The available options are determined by the selected encoding format.

3. Configure the options in the **Preprocessing** tab to adjust the settings for filtering, surround attenuation, and other preprocessing parameters applicable to the selected encoding format.

4. Configure timecode embedding in the **Timecode** tab.

   a) Select a timecode format and starting time.

   b) Specify a timecode range when encoding just a portion of the input file, also defining any prepend or append silence to be added to the beginning or to the end of the file.

Related information

Dolby Digital encoder settings on page 67
Dolby Digital Plus encoder settings on page 52
Dolby TrueHD encoder settings on page 38
MLP encoder settings on page 61

4.1.7 Ensuring the encoded audio aligns with video

When encoding only a portion of the source files, you can specify the time range for encoding to ensure that the encoded audio aligns with the encoded video.

About this task

You can specify time range relative to embedded timecode, or relative to file position.

Note: When mastering for Dolby Atmos, we recommend adding audio preroll greater than four seconds to the audio before recording master files. When .atmos master files are loaded into Dolby Media Encoder, you can set the audio encoding start point to align with the video start frame at the first video frame for encoding, sometimes called “first frame of action” or “first frame of picture.”

Procedure

1. On the **Encoder Setup** tab of a job editor window, click the **Timecode** tab.

2. From the **Timecode Frame Rate** drop-down menu, select a frame rate.

   The default value is 23.976fps.

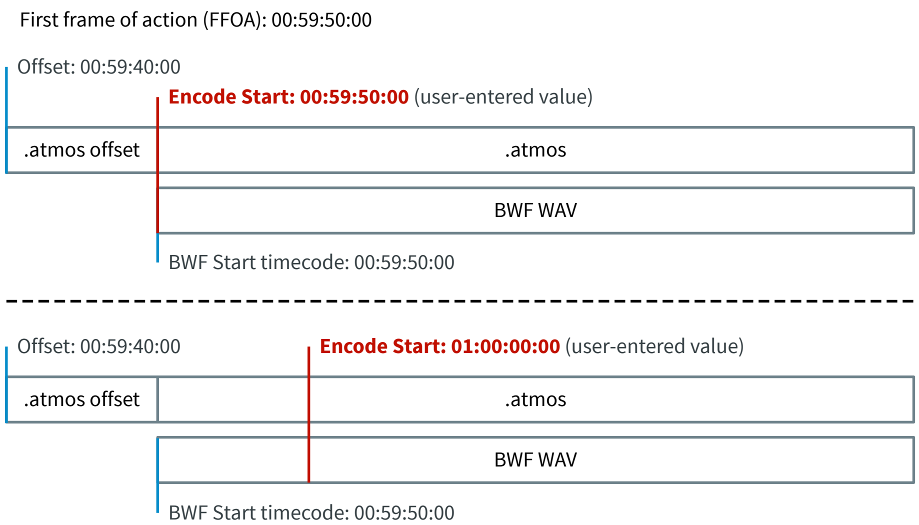
   Loading a Dolby Atmos master file as source will automatically populate this value with the frame rate of the .atmos file.

3. Enable the time range setup options by clicking **Specify Time Range to Encode**.
4. From the **Time Range Relative To** drop-down menu, select one of the options.

- **Embedded Timecode**

  This option takes into consideration both Dolby Atmos master file offset and BWF timestamp. The behavior is optimized for scenarios where the BWF timestamp is equal to the first frame of action of all input assets.

  *Figure 4: Example of **Embedded Timecode** encode start settings*

  The encoder applies a starting timecode of 00:00:00:00 for files without embedded timecode. In such scenarios the values of the **Encode Start** and **Encode End** fields only affect Dolby Atmos presentations.

- **File Position**

  This option makes all timecode adjustments relative to the start of the input files. Dolby Media Encoder ignores any Dolby Atmos master file offset or BWF timestamp and assumes the input files start at absolute 0.

  Note: Specifying an encode start time of 00:00:01:00 instructs Dolby Media Encoder to start encoding from the first minute of the input files.

5. In the **Encode Start** field, enter the timecode at which you wish to begin encoding.

   For Dolby Atmos material, this timecode must be the same or later than the **Dolby Atmos Master Start Time** displayed at the top of the tab.

6. Optional: Modify the **Encode End** time.

   The encode end time is set by default to the end of the input file, but you can choose to trim a portion at the end of the file.

7. Optional: Specify a number of silent frames or milliseconds to be added at the beginning of the encoded file in the **Prepend Silence** field.

   The available input options depend on the value of the **Timecode Frame Rate**.

8. Optional: Specify a number of silent frames or milliseconds to be added at the end of the encoded file in the **Append Silence** field.

   The available input options depend on the value of the **Timecode Frame Rate**.

9. Optional: Specify the output file start timecode and frame rate.

   a) From the **Specify New Output Timecode** drop-down menu, select the appropriate option.

   b) In the **Output File Start Timecode** field, enter a timecode value.

c) From the drop-down menu at the right of the **Output File Start Timecode** field, select an appropriate output frame rate.

4.1.8 Entering user information for encoding jobs

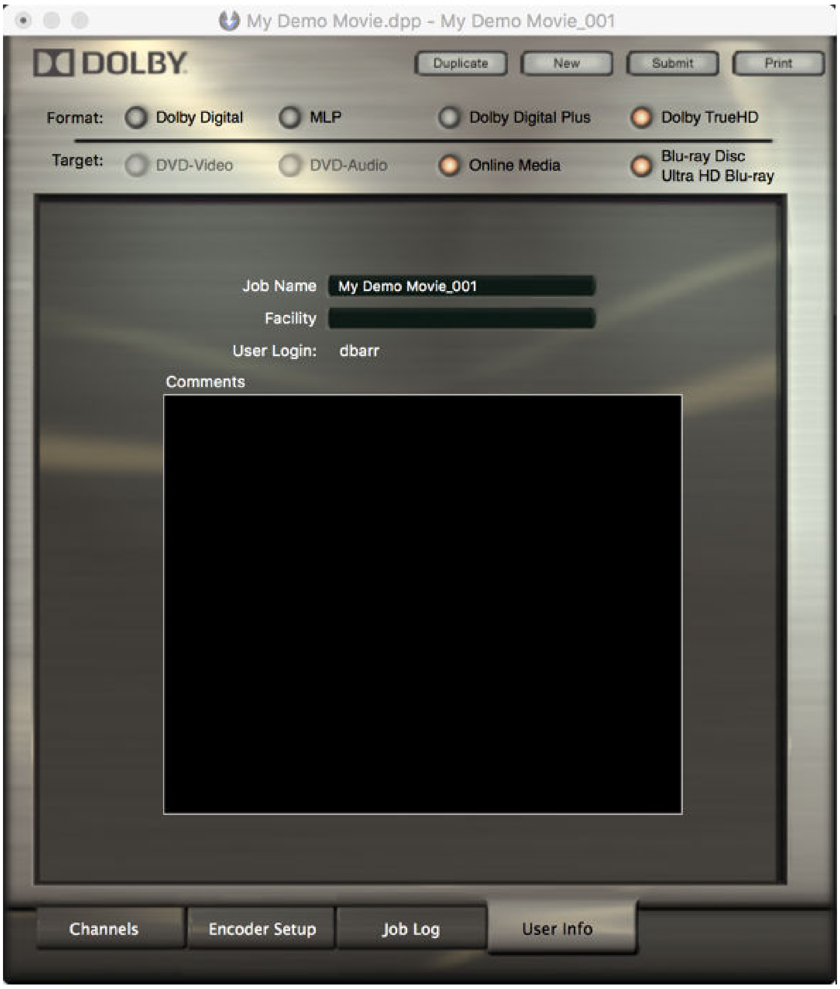
You can provide user information for each encoding job you submit. User information includes a job name, facility, and comments.

About this task

Dolby Media Encoder automatically populates the **Job Name** field. User Login data is taken from the active computer user. Other user information fields are optional.

Procedure

1. In the job editor window, click the **User Info** tab.



2. In the **Job Name** and **Facility** fields, type in new information
3. In the **Comments** section, add additional comments.
4. Save your changes by selecting **File** > **Save** on the Dolby Media Encoder menu bar.

4.1.9 Submitting an encoding job

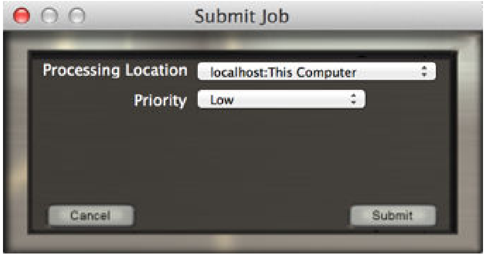
Encoding jobs are submitted from the Dolby Media Encoder Client to a local of remote instance of the Dolby Media Encoder Server application.

Procedure

1. At the top right of the job editor window, click **Submit**.

   Tip: If the current project has not been saved you will be prompted to create a project document. Click **Yes** to continue.

   The system displays the **Submit Job** dialog box.

2. From the **Processing location** drop-down menu, select a server.

   Dolby Media Encoder Client automatically sets this field to the last used location.

3. From the **Priority** drop-down menu, select an encoding priority.

   Priority affects the order in which encoding jobs are handled in the server encoding queue.

4. Click **Submit**.

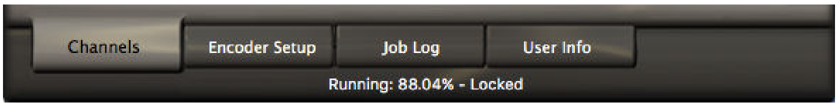
   If your network setup requires authentication, the system displays the required login dialog boxes to access source and destination locations.

4.1.10 Monitoring encoding job progress

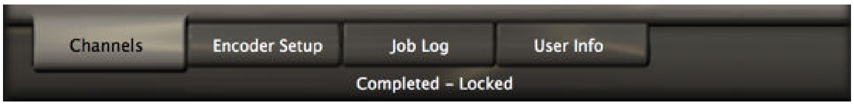
Dolby Media Encoder reports encoding progress at the bottom of the job editor window and in the job entry in the **job list** window.

- View the job progress at the bottom of the job editor window.

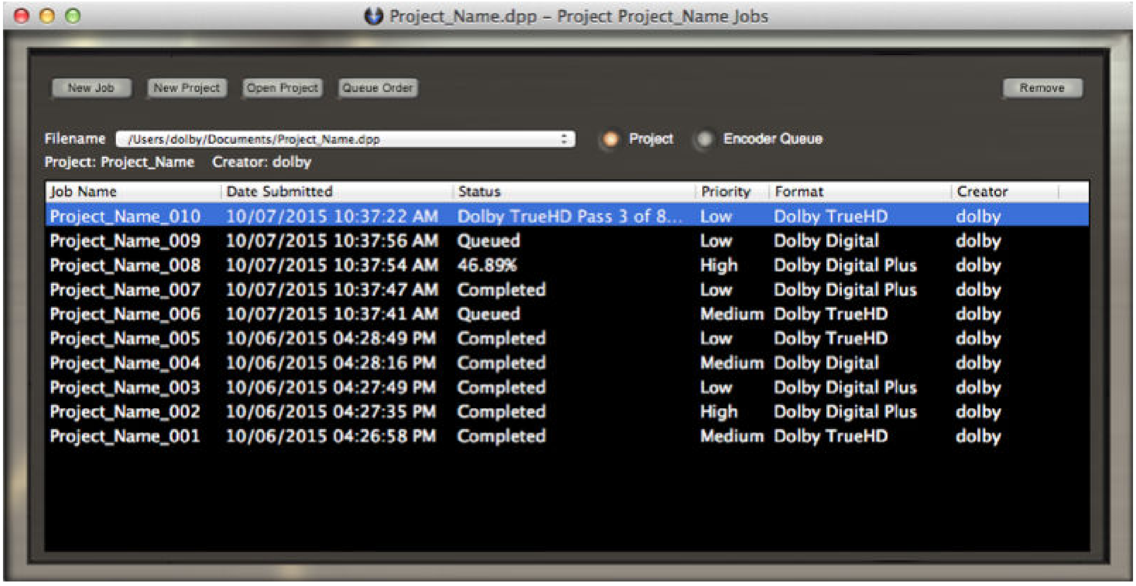
  Submitted jobs display the **Locked** status.

  When encoding finishes, the status changes to **Completed - Locked**.

- View the status of all active encoding jobs from the project **job list** window.



    - Clicking the heading of any column sorts the list by that field.
    - Clicking the **Queue Order** at the top of the window restores the initial sorting order.
- Access detailed information on complete encoding jobs from the **Job Log** tab in the job editor window.

4.2 Creating a Dolby TrueHD bitstream

You can create a Dolby TrueHD bitstream from a Dolby Atmos master file set and from independent channel-based source files.

Prerequisites

Verify that an iLok USB key is inserted into the machine running Dolby Media Encoder Server.

Procedure

1. Launch Dolby Media Encoder.
2. Select an existing project, or create a new project.
3. Create a new encoding job, or select an existing job from the project **job list**.
4. In the **Target** section of the **job editor** window, select **Blu-ray Disc and Ultra HD Blu-ray** .

   The options in the **Target** section are automatically set to **Online Media** and **Blu-ray Disc and Ultra HD Blu-ray** . Dolby Media Encoder produces a single encoded file compatible with both media targets.
5. If you want to encode from a Dolby Atmos master file set, activate the **Include Dolby Atmos** button at the top of the **Channels** tab of the job editor window.
6. Select the appropriate source options under the **8-Channel Presentation** drop-down menu.

   Dolby TrueHD bitstreams include 8-channel, 6-channel, and 2-channel presentations.

   You can encode independent presentations from unique source files, as long as the program material complexity does not cause the data rate or file size to exceed format limitations.

- To encode from a Dolby Atmos master file set to a Dolby TrueHD bitstream with Dolby Atmos, from the **8-Channel Presentation** drop-down menu, select **Downmix from Dolby Atmos** .
- To encode from a Dolby Atmos master file set to a channel-based 7.1 Dolby TrueHD bitstream, from the **8-Channel Presentation** drop-down menu, select **Dolby Atmos render to 7.1 (Lrs, Rrs)**.
- To encode from independent channel-based presentations, select the appropriate configuration from the **8-Channel Presentation** drop-down menu.
- If you are encoding from independent channel-based, you can additionally select to copy sources from another presentation by selecting **Copy of 6-channel** or **Copy of 2-channel** options.

7. Select the appropriate source options under the remaining **6-Channel Presentation** and **2-Channel Presentation** drop-down menus.
   - If you wish to have these presentations as downmixes of the 8-channel presentation, select **Downmix of 8-channel**.
   - A 2-channel presentation can alternatively be selected as a downmix of the 6-channel presentation, by selecting **Downmix of 6-channel**.
   - If you wish to provide independent source files for these presentations, select the appropriate configuration from the presentation drop-down menu.
   - If you are encoding from independent channel-based sources, you can choose to copy sources from the 2-channel presentation by selecting the **Copy of 2-channel** option.
8. If you are encoding from a Dolby Atmos Master file set, load the master file.
   a) Click (browse) to the right of the **Include Dolby Atmos** field.
   b) Locate and select your .atmos master file.
   c) To add the master file to the presentation, click **Open**.
   Alternatively, you can also drag and drop an .atmos file into the **Include Dolby Atmos** field.

   You can find information about the loaded Dolby Atmos master file set by clicking **Details** at the left of the **Include Dolby Atmos** field. The following information is shown:
   - Presentation type
   - Timecode
   - Encryption
   - Dialnorm
   - PCM file
   - Metadata file
   - Bed channel list
   - Total bed channels
   - Total objects
9. Add any required channel-based PCM source files for each channel-based presentation.
   Dolby TrueHD supports channel-based encoding at 44.1, 48, 88.2, 96, 176.4, and 192 kHz.
   Presentations require at least one input channel to be loaded. Dolby Media Encoder pads the remaining unloaded channels with silence.

Note: When encoding with both an .atmos master and PCM source files into multiple independent presentations, broadcast wave files (BWF) with embedded timecodes are automatically aligned with the .atmos master source file. PCM source signals without embedded timecodes must be trimmed to the required start and end time points.

10. In the **Destination Path** field, verify the destination path.
11. On the **Encoder Setup** tab, further configure the encoder settings, if needed.

    The **Encoder Settings** tab presents different options depending on whether the **Measure Loudness** option is enabled or disabled.

    Encoding from a Dolby Atmos master file set enables some additional configuration options in the **Encoder Setup** > **Preprocessing** tab.

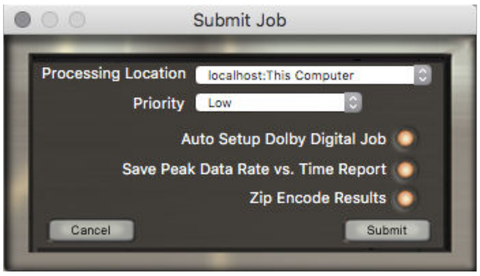
    a) Select the number of clustered signals to be output by spatial coding in the **Number of Elements** drop-down menu.

       If you are encoding into a bitstream with Dolby Atmos, the number of clustered signals to be output by spatial coding is hard-coded to 16.

    b) Enable the **Optimize Data Rate** option to apply an advanced data rate reduction algorithm to the intermediate signals used in the encoder signal path. This can reduce peak and average data rates by up to 30-40%.
12. On the **User Info** tab, enter the user information for the job.
13. Click **Submit**.

    The Blu-ray Disc specification requires Dolby TrueHD bitstreams to include an accompanying Dolby Digital soundtrack. The **Submit Job** dialog box for Dolby TrueHD has additional options to simplify the encoding workflow.

    a) To automatically launch a Dolby Digital job editor with settings copied over from your Dolby TrueHD job, activate the **Auto Setup Dolby Digital** button.



    b) Optional: Include a **Peak data rate vs. time report** together with your encode job output file. The report provides a breakdown of peak data rate for each 1-second interval of the encoded Dolby TrueHD bitstream, and is stored as a comma-separated value file with .mlpbr extension.

    c) Optional: Compress your encode results into a .zip file by activating the **Zip Encode Results** button.

Results

When the encoding finishes, the job status changes to **Completed - Locked**. An .mlp file is now available at the destination path.

Note: Dolby Media Encoder can process up to four simultaneous channel-based Dolby TrueHD encoding jobs, and two simultaneous encoding jobs for Dolby TrueHD bitstreams with Dolby Atmos.

Dolby TrueHD encoding is a multipass process. Dolby Media Encoder provides additional status information about pass states:

- Spatial coding (Dolby Atmos bitstreams only)
- Matrixing (Dolby Atmos bitstreams only)
- Analyze
- Encode
- Minimize data rate
- Verify

Related information

Dolby TrueHD encoder setup options on page 37

4.3 Creating a Dolby Digital Plus bitstream

You can create a Dolby Digital Plus bitstream from a Dolby Atmos master file set and from independent channel-based source files.

Prerequisites

Verify that an iLok USB key is inserted into the machine running Dolby Media Encoder Server.

About this task

With Dolby Media Encoder, you can also encode Dolby Digital Plus Blu-ray Disc secondary audio tracks.

Procedure

1. Launch Dolby Media Encoder.
2. Select an existing project, or create a new project.
3. Create a new encoding job, or select an existing job from the project **job list**.
4. In the **Format** section of the **job editor** window, select **Dolby Digital Plus** .
5. In the **Target** section of the **job editor**, select an appropriate media target.
   - **Online Media**
   - **Blu-ray Disc and Ultra HD Blu-ray**
6. Select the appropriate channel configuration under the **Channel Config** drop-down menu.
   - To encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos, select **Dolby Atmos Bitstream**.
   - To encode from a Dolby Atmos master file set to a 7.1 channel-based Dolby Digital Plus bitstream, select **Dolby Atmos render to 7.1 (Lrs, Rrs)**.

     Note: This option is only available for the Blu-ray Disc and Ultra HD Blu-ray media target.
   - To encode from a Dolby Atmos master file set to a 5.1 channel-based Dolby Digital Plus bitstream, select **Dolby Atmos render to 5.1 (Ls, Rs)**.

     Note: This option is only available for the Online Media media target.
   - To encode from independent channel-based presentations, select the appropriate configuration.
   - To encode for a Blu-ray Disc secondary audio track, select the **Blu-ray Secondary Audio** option at the bottom of the **Channels** tab, and select the appropriate configuration on the **Channel Config** drop-down menu.

> **Note:** This option is only available for the Blu-ray Disc and Ultra HD Blu-ray media target.

7. Load the source files.
   - Loading an .atmos master file effectively loads all three master source files.
   - Channel-based presentations require at least one input channel to be loaded. Dolby Media Encoder pads the remaining unloaded channels with silence.
8. In the **Destination Path** field, verify the destination path.
9. Select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital bitstream, choosing one of the options under **5.1 Downmix Type** or **5.1 Core Type**.

   When encoding from a .atmos or .wav master file, the 5.1 core can be created in two ways, based on the following scenarios:

   a) To create the 5.1 core by first rendering the Dolby Atmos audio to 7.1 channel configuration and then downmixing the four surround channels into two surround channels based on the Dolby Pro Logic IIx downmix equations, select **5.1 Dolby PLIIx**.

   b) To create the 5.1 core based on the `warpMode` metadata value found in the master file, select **5.1 Standard / Auto**. If no `warpMode` metadata is found in the master file, `warping` mode will be selected as default.
10. On the **Encoder Setup** tab, further configure the encoder settings, if needed.
11. On the **User Info** tab, enter user information for the job.
12. Click **Submit**.

Results
When the encoding finishes, the job status changes to **Completed - Locked**. An .ec3 file (for Online Media jobs) or .eb3 file (for Blu-ray Disc and Ultra HD Blu-ray jobs) is now available at the destination path.

Related information

Dolby Digital Plus encoder setup options on page 50

4.3.1 Warp mode

This field controls how the 5.1 Core is generated when encoding from Dolby Atmos Master files.

This field is only valid for speaker configurations where the speakers on the listener plane that are not on the front are surround speakers (Ls/Rs or Ls/Rs/Cs), such as in a standard 5.1 or 5.1.2 speaker layout. The value of this field is interpreted as shown in the following table:

warpMode	Description
"normal"	Normal rendering (no object position adjustment applied).
"warping"	Object position is warped prior to rendering. Warped position scales the y-coordinate by applying a factor of 2 (y(warped) = 2*y). This is Auto (Lo/Ro) mode in Dolby Media Encoder.
"ProLogicIIx"	Normal rendering to a 7.1 configuration, followed by a 7.1-to-5.1 Dolby Pro Logic IIx-style downmix of the surround channels.
"LoRo"	Normal rendering to a 7.1 configuration, followed by a 7.1-to-5.1 LoRo-style downmix of the surround channels.

4.4 Creating an MLP bitstream

You can create an MLP bitstream from independent channel-based source files.

Prerequisites
Verify that an iLok USB key is inserted into the machine running Dolby Media Encoder Server.

Procedure

1. Launch Dolby Media Encoder.
2. Select an existing project, or create a new project.
3. Create a new encoding job, or select an existing job from the project **job list**.
4. In the **Format** section of the **job editor** window, select **MLP**.

   The MLP format only supports the **DVD-Audio** media target.
5. Select a channel configuration on the **6-Channel Presentation** drop-down menu on the **Channels** tab.
6. Load the source files.

   Presentations require at least one input channel to be loaded. Dolby Media Encoder pads the remaining unloaded channels with silence.
7. Leave the configuration of the **2-Channel Downmix** drop-down menu at its default settings, unless Dolby support recommends different settings.
8. In the **Destination Path** field, verify the destination path.
9. On the **Encoder Setup** tab, further configure the encoder settings, if needed.
10. On the **User Info** tab, enter the user information for the job.
11. Click **Submit**.

Results
When encoding finishes, the job status changes to **Completed - Locked**. An .mlp file is now available at the destination path.

Related information

MLP encoder setup options on page 60

4.5 Creating a Dolby Digital bitstream

You can create a Dolby Digital bitstream from a Dolby Atmos master file set and from independent channel-based source files.

Prerequisites
Verify that an iLok USB key is inserted into the machine running Dolby Media Encoder Server.

About this task
When encoding from a Dolby Atmos master file set, the Dolby Digital bitstream does not carry Dolby Atmos metadata.

Procedure

1. Launch Dolby Media Encoder.
2. Select an existing project, or create a new project.
3. Create a new encoding job, or select an existing job from the project **job list**.
4. In the **Format** section of the job editor window, select **Dolby Digital** .
5. In the **Target** section of the job editor, select an appropriate media target.
   - **DVD-Video**
   - **Blu-ray Disc and Ultra HD Blu-ray**
   - Both **DVD-Video** and **Blu-ray Disc and Ultra HD Blu-ray**
6. On the **Channel Config** drop-down menu of the **Channels** tab, select a channel configuration.

   Note: To encode from a Dolby Atmos master file set, select **Dolby Atmos render to 5.1 (Ls, Rs)**.
7. Load source files.

   Presentations require at least one input channel to be loaded. Dolby Media Encoder pads the remaining unloaded channels with silence.
8. In the **Destination Path** field, verify the destination path.
9. Select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital bitstream by selecting **5.1 Downmix Type**.

   When encoding from a .atmos or .wav master file, the 5.1 core can be created in two ways based on the following scenarios:

   a) To create the 5.1 core by first rendering the Dolby Atmos audio to 7.1 channel configuration and then downmixing the four surround channels into two surround channels based on the Dolby Pro Logic IIx downmix equations, select **5.1 Dolby PLIIx**.

   The coefficients for a Dolby Pro Logic IIx downmix from 7.1 to 5.1 are:
   Ls = Lss + (-1.2 dB x Lrs) + (-6.2 dB x Rrs)
   Rs = Rss + (-6.2 dB x Lrs) + (-1.2 dB x Rrs)

   b) To create the 5.1 core based on the `warpMode` metadata value found in the master file, select **5.1 Standard / Auto**. If no `warpMode` metadata is found in the master file, `warping` mode will be selected as default.
10. On the **Encoder Setup** tab, further configure the encoder settings, if needed.
11. On the **User Info** tab, enter the user information for the job.
12. Click **Submit**.

Results

When the encoding finishes, the job status changes to **Completed - Locked**. An .ac3 file is now available at the destination path.

Related information

Dolby Digital encoder setup options on page 65
Warp mode on page 33

4.6 Troubleshooting encoding problems

Dolby Media Encoder includes an log feature that can assist in troubleshooting unsuccessful encoding jobs.

Procedure

1. Access the job log on the **Job Log** tab of the **job editor** window.
2. Take note of the errors listed in the job log.
3. Duplicate the failed job and correct any error conditions noted in the previous job log.
4. Submit the new job.

5 Dolby TrueHD encoder setup options

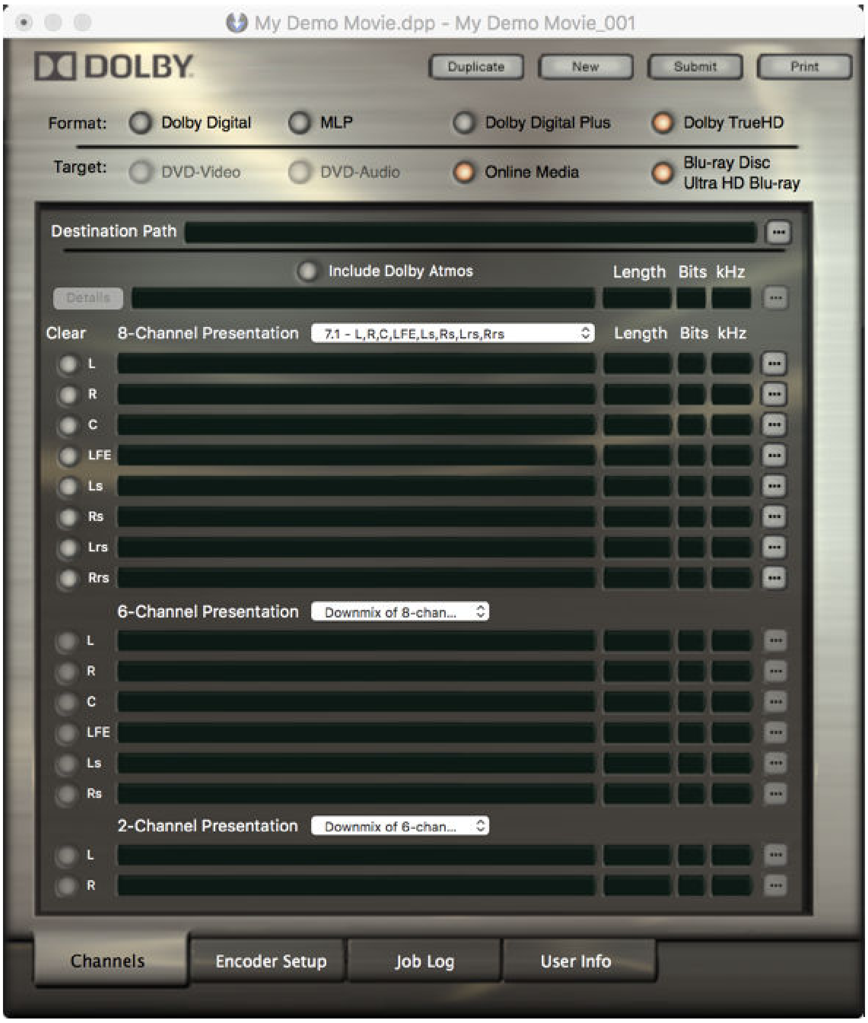
The selected format and targets of an encoding job determine its available encoding configurations and settings.

- Dolby TrueHD encoding configurations
- Dolby TrueHD encoder settings
- Dolby TrueHD downmix settings
- Dolby TrueHD preprocessing settings
- Dolby TrueHD timecode settings

5.1 Dolby TrueHD encoding configurations

For Dolby TrueHD encoding jobs, Dolby Media Encoder presents channel configuration and source-file options for Dolby Atmos and eight-, six-, and two-channel presentations.

*Figure 5: Dolby TrueHD **Channels** tab*



Dolby TrueHD presentations can be:

- Independent, encoded from independent source files
- Downmixed from an independent presentation with more channels

- Copied from an independent presentation

Table 3: Dolby TrueHD 8-channel presentations

Encoding configuration	Comments
Copy of 2-channel	Only available for channel-based bitstreams
Copy of 6-channel	Only available for channel-based bitstreams
7.1 - L, R, C, LFE, Ls, Rs, Lrs, Rrs	Default configuration
Downmix from Dolby Atmos	Default configuration with Dolby Atmos source files
Dolby Atmos render to 7.1 (Lrs, Rrs)	

Table 4: Dolby TrueHD 6-channel presentations

Encoding configuration	Comments
Copy of 2-channel	Only available for channel-based bitstreams
Downmix of 8-channel	Default configuration
5.1 - L, R, C, LFE, Ls, Rs	
3.0 - L, C, R	
3.1 - L, C, R, LFE	
4.0 - L, R, Ls, Rs	
5.0 - L, R, C, Ls, Rs	

Table 5: Dolby TrueHD 2-channel presentations

Encoding configuration	Comments
2-channel	
Downmix of 6-channel	Default configuration
Downmix of 8-channel	

Related information

Selecting encoding configuration and loading source files on page 21

5.2 Dolby TrueHD encoder settings

On the **Encoder Settings** tab under the **job editor Encoder Setup** tab, you can modify default metadata parameters and other encoding settings, including Loudness measurement.

The **Encoder Settings** tab presents different options depending on whether the **Measure Loudness** option is enabled or disabled.

*Figure 6: Dolby TrueHD **Encoder Settings** tab with Loudness measurement disabled*

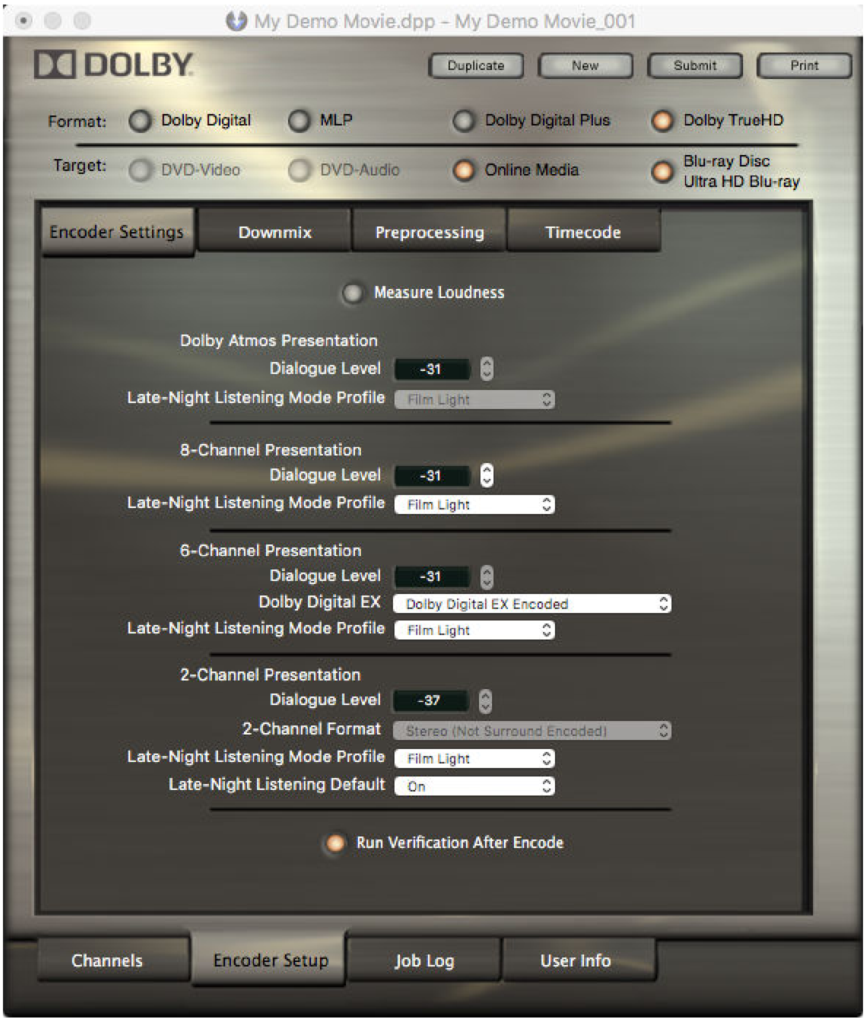
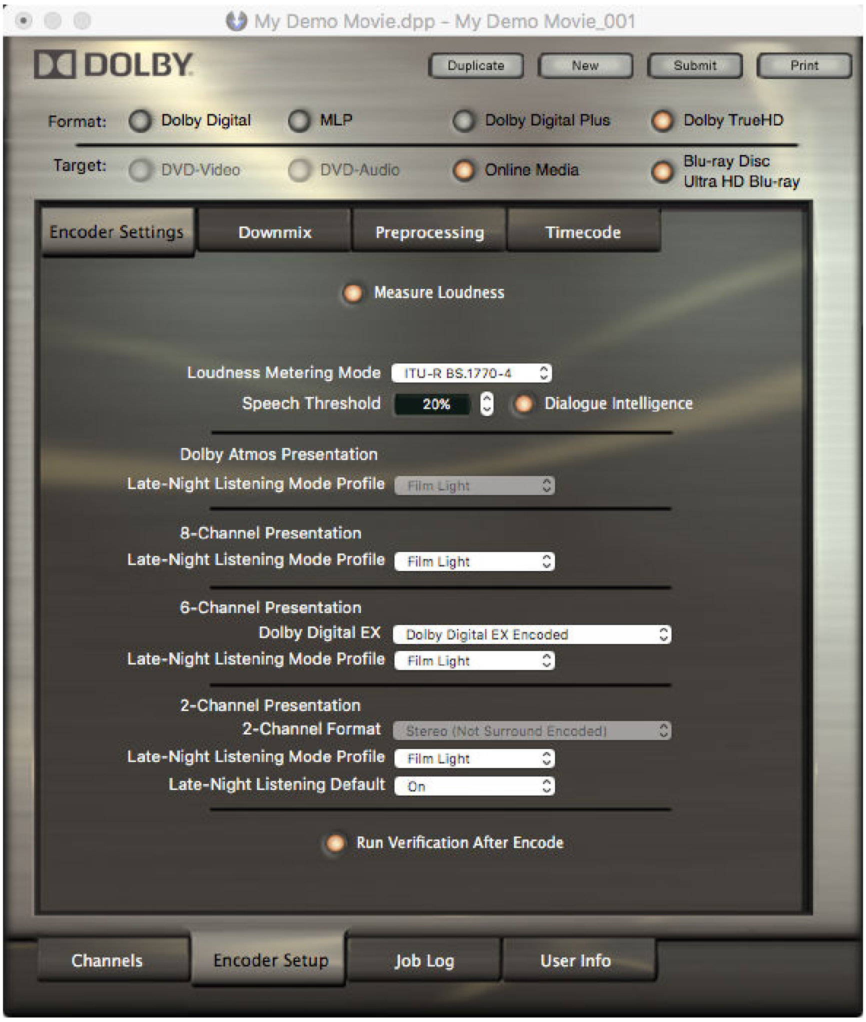


*Figure 7: Dolby TrueHD **Encoder Settings** tab with Loudness measurement enabled*



Dolby TrueHD encoder settings are configured separately for each presentation. Copied presentations have no independent settings options.

Note: The presence of multiple independent presentations increases file size and data rate.

Loudness measurement options

Setting	Range
Loudness Metering Mode	ITU-R BS. 1770-4
	ITU-R BS. 1770-3
	ITU-R BS. 1770-2
	ITU-R BS. 1770-1
	Leq(A)
Dialogue Intelligence	Enabled (default)
	Disabled
Speech Threshold *	0% to 100% (default 20%)

* This setting is only active when **Dialogue Intelligence** is enabled.

- The **Loudness Metering Mode** setting specifies the weighting (A-weighting or K-weighting) used to estimate perceived loudness and whether dialogue should serve as the anchor element for measuring loudness. These modes include ITU standards and equivalent continuous A-weighted sound pressure level.

- The **Dialogue Intelligence** parameter enables or disables a feature that identifies and analyzes dialogue segments as a basis for speech gating and loudness measurement.
- The **Speech Threshold** setting defines the minimum amount of speech a program must contain in order for dialogue to be used as the anchor element for loudness measurements. When Dialogue Intelligence detects that the Speech Threshold is exceeded, the Loudness Metering Mode reverts to ITU-R BS. 1770-1 for the content identified as speech, that is, the silence gate and relative gate used in ITU-R BS. 1770-2, -3, and -4 are not used.

Note: These options are only available if the **Measure Loudness** option is enabled.

Dolby Atmos Presentation

Setting	Range
Dialogue level *	-1 to -31 (default -31)
Late-Night Listening Mode Profile	Film Standard
	Film Light (default)
	Music Standard
	Music Light
	Speech

* Not available if the **Measure Loudness** option is enabled.

8-Channel Presentation

Setting	Range
Dialogue level *	-1 to -31 (default -31)
Late-Night Listening Mode Profile	Film Standard
	Film Light (default)
	Music Standard
	Music Light
	Speech

* Not available if the **Measure Loudness** option is enabled.

6-Channel Presentation

Setting	Range
Dialogue level *	-1 to -31 (default -31)
Dolby Digital Surround EX	Dolby Digital EX Encoded
	Not Indicated
	Not Dolby Digital EX Encoded
Late-Night Listening Mode Profile	Film Standard
	Film Light (default)
	Music Standard
	Music Light
	Speech

Setting	Range

* Not available if the **Measure Loudness** option is enabled.

2-Channel Presentation

Setting	Range
Dialogue level *	-1 to -31 (default -31), or -37 if the presentation is a downmix
2-Channel Format	Dolby Surround Encoded
	Mono
	Stereo (Not Surround Encoded)
	Dolby Headphone Encoded
Late-Night Listening Mode Profile	Film Standard
	Film Light (default)
	Music Standard
	Music Light
	Speech
Late-Night Listening Mode Default †	On/Off

* Not available if the **Measure Loudness** option is enabled.

† Available when the presentation is a downmix of 6- or 8-channel presentations

Related information

Configuring additional encoder settings on page 23
Dynamic range control on page 82

5.3 Dolby TrueHD downmix settings

On the **Downmix** tab under the **job editor Encoder Setup** tab, you can modify the default metadata used for instructing a decoder on how to downmix its output.

*Figure 8: Dolby TrueHD **Downmix** tab*

For **8- to 6-Channel Presentation Downmix**, you can select:

- Dolby Pro Logic IIx
- Standard
- Custom

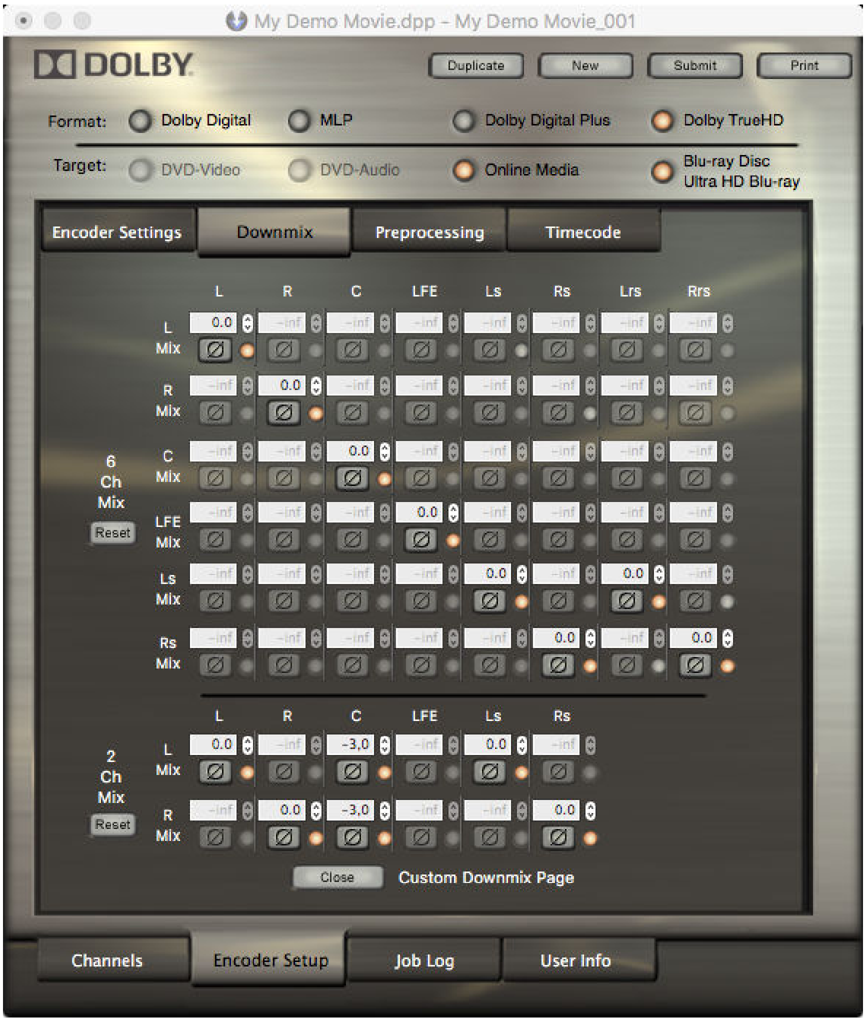
For **6- to 2-Channel Presentation Downmix**, you can select:

- Standard
- Custom

Custom downmix

If you select a **Custom** downmix option, you must click **Open** at the bottom of the **Downmix** tab to open the **Custom Downmix** page.

*Figure 9: Dolby TrueHD **Custom Downmix** page*



On the **Custom Downmix** page, you can select the channels you wish to include, as well as their levels and phase settings.

Each channel includes a **Phase** button  to invert the phase of the signal.

Related information

Downmixing in Dolby systems on page 83

5.4 Dolby TrueHD preprocessing settings

On the **Preprocessing** tab under the **job editor Encode Setup** tab, you can modify the default values for several optional processes that can be applied to source files as part of the encoding process.

*Figure 10: Dolby TrueHD **Preprocessing** tab*

Preprocessing settings for channel presentations

Channel presentations share similar preprocessing options:

Surround Channel 3 dB Attenuation

Applies a 3 dB attenuation to surround channel source files before encoding. This option does not apply to the 2-channel presentation.

Note: Surround channel attenuation is appropriate when encoding a multichannel soundtrack created in a room with film-style calibration for consumer home theater playback. Channel-based cinema soundtrack surround channels are traditionally recorded 3 dB louder relative to the front channels, which requires that the surround loudspeakers are set 3 dB lower than the front (screen) channels. Home theater surround speakers are calibrated differently, with all speakers playing back at the same level.

Apply Apodization Filter

The apodization filter assists in minimizing the ringing effect that can occur in some analog-to-digital and digital-to-analog filters. Apodization filters can aid in decoding efficiency and improve the response of the audio through many A/D converters. The apodization filter is applicable only to source files with a sampling rate of at least 88.2 kHz.

Apply Advanced Upsampling

Applies an upsampler to all .wav files loaded into the presentation fields. The option is available for 44.1 or 48 kHz audio sample rates, when encoding from Dolby TrueHD without

Dolby Atmos. When selected, 44.1 kHz audio is upsampled to 88.2 kHz, and 48 kHz audio is upsampled to 96 kHz.

Optimize Data Rate

Enabling this option reduces peak data rates by 25-35% and average data rates by 30-40%.

- For 44.1 or 48 kHz audio sample rates, the encoder applies an advanced data rate reduction algorithm to intermediate signals in 8-channel and 6-channel presentations, and applies noise shaping to 2-channel presentations.
- For audio sample rates above 48 kHz, the encoder applies noise shaping to all channel presentations.

Note: Noise shaping reduces the entropy of the low-level noise in the least-significant bits of a 24-bit source file. Minimizing this noise can aid in coding efficiency without affecting audio quality. Noise shaping has no effect on 16-bit source files.

Preprocessing settings for Dolby Atmos master file sets

Encoding from a Dolby Atmos master file set enables some additional configuration options.

Number of Elements

Sets the number of clustered signals plus bed signals of Dolby Atmos material to be output by spatial coding.

16

Use this setting if you have no data space considerations and want the absolute best quality. This setting is hard-coded when rendering audio from Dolby Atmos master files.

14

Use this setting if space allows it and an extra measure of quality is desired.

12 (default)

Use this setting for Blu-ray Discs with space considerations, as it provides high quality at an efficient average and peak data rate.

Optimize data rate

Applies an advanced data rate reduction algorithm to intermediate signals used in the encoder signal path when encoding from a Dolby Atmos master file set to Dolby TrueHD. This can reduce the peak and average data rates by 25-40% for typical content.

5.5 Dolby TrueHD timecode settings

The **Timecode** tab under the **job editor Encoder Setup** tab provides the same display and options for Dolby TrueHD, Dolby Digital Plus, and Dolby Digital encoding jobs.

*Figure 11: Dolby TrueHD **Timecode** tab*

The **Timecode** tab provides options to enable or disable inclusion of SMPTE time stamps in the output bitstream, as well as options to choose the timecode format and starting time for any present embedded timecode.

Note: Time stamps are displayed in hours:minutes:seconds:milliseconds format.

Dolby Atmos Master Start Time

This option is visible when encoding from a Dolby Atmos master file set with embedded timecode; it displays the start time of the Dolby Atmos input content.

Timecode Frame Rate

This option defines the frame rate or real-time value of the encoding.

Timecode Frame Rate value	Comments
23.976 (default)	Frames-per-second
24	Frames-per-second
29.97	Frames-per-second
29.97 df	Drop-frame Frames-per-second
30	Frames-per-second

Timecode Frame Rate value	Comments
Real Time (ms)	Use this option if you need to edit the encode range in exact milliseconds and the Dolby Atmos file set was recorded in a “real time” SMPTE frame rate.
NTSC -0.1% (ms)	Use this option if you need to edit the encode range in NTSC milliseconds and the Dolby Atmos master file set was recorded at a slower NTSC frame rate. This option treats the entered time as slower than real time, by a factor of 1.001 of real time, or -0.1%. This makes NTSC seconds last longer than real time seconds, being 48048 samples long instead of 48000. *

* Refer to *SMPTE ST 12-1:2014, Time and Control Code* for further information on NTSC versus Real Time frame rates.

Note: Modifying the **Timecode Frame Rate** resets the **Encode Start** and **Encode End** values to the whole frame (or millisecond) value closest to the start and end of the input file. We recommend setting the frame rate for the encoding job before modifying the **Encode Start** and **Encode End** points.

Specify Time Range to Encode

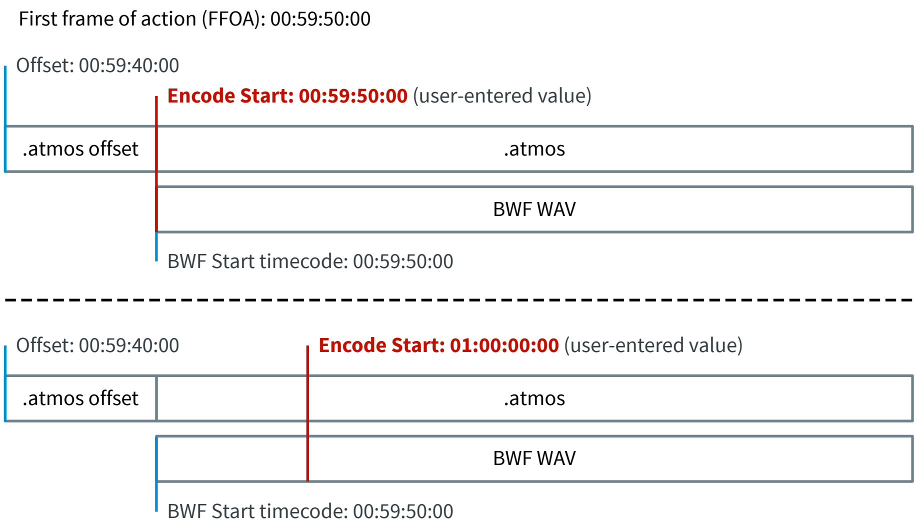
By activating this option you can specify a range that is a portion of the input files to encode.
This range is by default empty.

Time Range Relative To

In this menu, you can select whether the specified position is relative to the **Embedded Timecode** position or to the **File Position**, considering the start of the file as the 00:00:00:00 time stamp.

- The **Embedded Timecode** option takes into consideration both Dolby Atmos master file offset and BWF timestamp. The behavior is optimized for scenarios where the BWF timestamp is equal to the first frame of action of all input assets.

  Figure 12: Example of Embedded Timecode encode start settings

  The encoder applies a starting timecode of 00:00:00:00 for files without embedded timecode. In such scenarios the values of the **Encode Start** and **Encode End** fields only affect Dolby Atmos presentations.

- The **File Position** option makes all timecode adjustments relative to the start of the input files. Dolby Media Encoder ignores any Dolby Atmos master file offset or BWF timestamp and assumes the input files start at absolute 0.

Note: Specifying an encode start time of 00:00:01:00 instructs Dolby Media Encoder to start encoding from the first minute of the input files.

This option is by default set to **Embedded Timecode**. If the input files include no embedded timecode, the encode start reverts to the start of the file.

Encode Start
: Defines a valid time point from which to start encoding.

Encode End
: Defines a valid time point at which to end encoding.

Prepend Silence
: Use this option to set the duration of silence to add at the beginning of the encoded file. This field accepts valid integers for frames or milliseconds, based on the selected **Timecode Frame Rate**.

Append Silence
: Use this option to set the duration of silence to add at the end of the encoded file. This field accepts valid integers for frames or milliseconds, based on the selected **Timecode Frame Rate**.

Specify New Output Timecode
: Use this menu if you wish to define and include timecode in the output file. Dolby Media Encoder will remember your last selected setting for this menu. Options include:

- **Do not include timecode**
- **Use input timecode** (default)
- **Specify timecode**

Note: Specifying a timecode (**Specify timecode**) or using the input timecode (**Use input timecode**) can cause file incompatibility issues with third-party applications.

Output File Start Timecode
: The behavior of this option depends on the value of the **Specify New Output Timecode** option. By default this option is unavailable.
If **Use input timecode** is selected, this field displays the same value as the **Encode Start** field.
If **Specify timecode** is selected, the output file timecode can be changed to any positive timecode value, with a default value of 00:00:00:00.
You can find a frame rate selection drop-down menu at the right of the **Output File Start Timecode** option:

Output File Start Timecode frame rate value	Comments
23.976	Frames-per-second
24	Frames-per-second
29.97	Frames-per-second
29.97 df	Drop-frame Frames-per-second
30	Frames-per-second

6 Dolby Digital Plus encoder setup options

The selected format and targets of an encoding job determine its available encoding configurations and settings.

- Dolby Digital Plus encoding configurations
- Dolby Digital Plus encoder settings
- Dolby Digital Plus downmix settings
- Dolby Digital Plus preprocessing settings
- Dolby Digital Plus timecode settings

6.1 Dolby Digital Plus encoding configurations

For Dolby Digital Plus encoding jobs, Dolby Media Encoder presents different sets of channel configurations and source-file options, depending on media target and on whether the encode is meant for Blu-ray Disc secondary audio.

*Figure 13: Dolby Digital Plus **Channels** tab*

Table 6: Dolby Digital Plus encoding configurations

Encoding configuration	Media target	Comments
1.0 - Mono	Online Media and Blu-ray Disc secondary audio	
2.0 - Stereo	Online Media and Blu-ray Disc secondary audio	
2.0 - Lt/Rt	Online Media	Flagged as matrix-encoded Lt/Rt signal. Dolby Media Encoder only sets metadata in the encoded stream so that a decoder can identify how the tracks should be played back.
5.1 - L, R, C, LFE, Ls, Rs	Online Media and Blu-ray Disc secondary audio	Default configuration
5.1 Ex - L, R, C, LFE, Ls, Rs	Online Media	Flagged as including Dolby Surround EX content. Select this option for source files with additional matrix-encoded channels already mixed into the Ls and Rs inputs
5.1 - L, R, C, LFE, Ls, Rs (Lrs, Rrs)	Online Media	Lrs/Rrs source files are downmixed into the Ls/Rs channels before encoding, enabling additional downmix options. Select this option to produce Dolby Digital Plus 5.1-channel files from 7.1 source files.
3.0 - L, C, R	Online Media and Blu-ray Disc secondary audio	
3.1 - L, C, R, LFE	Online Media	
4.0 - L, C, R, S	Online Media and Blu-ray Disc secondary audio	
4.0 - L, R, Ls, Rs	Online Media and Blu-ray Disc secondary audio	
4.1 - L, R, Ls, Rs, LFE	Online Media and Blu-ray Disc secondary audio	
5.0 - L, R, C, Ls, Rs	Online Media and Blu-ray Disc secondary audio	
7.1 - L, R, C, LFE, Ls, Rs, Lrs, Rrs	Blu-ray Disc and Ultra HD Blu-ray	Default configuration
Dolby Atmos Bitstream		Encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos, enabling additional downmix options.

Table 6: Dolby Digital Plus encoding configurations (continued)

Encoding configuration	Media target	Comments
Dolby Atmos render to 5.1 (Ls, Rs)	Online Media	Encodes from a Dolby Atmos master file set to a 5.1 Dolby Digital Plus bitstream without Dolby Atmos, enabling additional downmix options. Select this option to create a 5.1-channel Dolby Digital Plus companion track for Blu-ray Disc.
Dolby Atmos render to 7.1 (Lrs, Rrs)	Blu-ray Disc and Ultra HD Blu-ray	Encodes from a Dolby Atmos master file set to a 7.1 Dolby Digital Plus bitstream without Dolby Atmos, enabling additional downmix options. Select this option to create a 7.1-channel Dolby Digital Plus companion track for Blu-ray Disc.

Some Dolby Digital Plus encoding configurations enable additional downmix options within the **Channels** tab.

Table 7: Dolby Digital Plus additional downmixing options

Downmix option	Comments
5.1 Dolby PLIIx	Default option
5.1 Standard (Auto)	

The coefficients for a Dolby Pro Logic IIx (PLIIx) downmix are:
Ls = Lss + (-1.2 dB x Lrs) + (-6.2 dB x Rrs)
Rs = Rss + (-6.2 dB x Lrs) + (-1.2 dB x Rrs)

The coefficients for a Standard (Auto) downmix are:
Ls = 0 dB x Lss + 0 dB x Lrs
Rs = 0 dB x Rss + 0 dB x Rrs

Related information

Selecting encoding configuration and loading source files on page 21
Downmixing in Dolby systems on page 83

6.2 Dolby Digital Plus encoder settings

On the **Encoder Settings** tab under the job editor **Encoder Setup** tab, you can modify default metadata parameters and other encoding settings, including Loudness measurement.

*Figure 14: Dolby Digital Plus **Encoder Settings** tab*

Loudness measurement options

Setting	Range
Loudness Metering Mode	ITU-R BS. 1770-4
	ITU-R BS. 1770-3
	ITU-R BS. 1770-2
	ITU-R BS. 1770-1
	Leq(A)
Dialogue Intelligence	Enabled (default)
	Disabled
Speech Threshold *	0% to 100% (default 20%)

* This setting is only active when **Dialogue Intelligence** is enabled.

- The **Loudness Metering Mode** setting specifies the weighting (A-weighting or K-weighting) used to estimate perceived loudness and whether dialogue should serve as the anchor element for measuring loudness. These modes include ITU standards and equivalent continuous A-weighted sound pressure level.
- The **Dialogue Intelligence** parameter enables or disables a feature that identifies and analyzes dialogue segments as a basis for speech gating and loudness measurement.

- The **Speech Threshold** setting defines the minimum amount of speech a program must contain in order for dialogue to be used as the anchor element for loudness measurements. When Dialogue Intelligence detects that the Speech Threshold is exceeded, the Loudness Metering Mode reverts to ITU-R BS. 1770-1 for the content identified as speech, that is, the silence gate and relative gate used in ITU-R BS. 1770-2, -3, and -4 are not used.

Note: Loudness measurement is always enabled for Dolby Digital Plus encoding jobs.

Other options

Setting	Range
Dialogue level	-1 to -31 (default -31)
Data rate	Based on the selected format, target, and number of input channels, Dolby Media Encoder automatically selects the most common data rate and provides a list of alternatives.
Bitstream Mode	Complete Main (default)
	Music & Effects
	Visually Impaired
	Hearing Impaired
	Dialogue
	Commentary
	Emergency
	Voice Over
Late-Night Listening Mode Profile, Line Mode	Film Light (default)
	Film Standard
	Music Standard
	Music Light
	Speech
Late-Night Listening Mode Profile, RF Mode	Film Light (default)
	Film Standard
	Music Standard
	Music Light
	Speech
Blu-ray Secondary Audio Settings, Primary Soundtrack Gain	-inf (infinity) or -50 to +12dB. Defaults to 0dB.
Blu-ray Secondary Audio Settings, Fade In/Out Frames	0 to 10. Defaults to 0.
Optional User Inserted ID Number	Defaults to 0 and accepts numbers with up to five digits.

Blu-ray Secondary Audio data rates

The following data rates are supported for Blu-ray Disc secondary audio:

- 160 – 256 kbps for 5.1 audio
- 96 – 256 kbps for 2.0 audio

- 32 – 256 kbps for 1.0 audio

Metadata processes

You can embed metadata for the secondary audio by choosing a fixed primary soundtrack gain and a number of fade-in/fade-out frame.

The primary soundtrack gain value tells the player how much attenuation to apply to the primary soundtrack when secondary audio is used. This is a static value that does not change throughout the encode. The range is –inf dB or –50 dB to +12 dB, in 1 dB increments.

The fade-in/fade-out frames value sets the number of frames over which secondary audio fades in and out. The range is 0 to 10.

Related information

[Configuring additional encoder settings](#) on page 23
[Dynamic range control](#) on page 82

6.3 Dolby Digital Plus downmix settings

On the **Downmix** tab under the job editor **Encoder Setup** tab you can modify the default metadata used for instructing a decoder on how to downmix its output.

*Figure 15: Dolby Digital Plus **Downmix** tab*

The options defined on the **Downmix** tab are different from the downmix options on the **Channels** tab and are meant to instruct decoders on how to downmix for two-channel output.

Setting	Range	Default value	Comments
Lt/Rt Center	-6 to +3 dB	-3 dB	Default mix level for center signals into L/R channels of Lt/Rt downmix
Lt/Rt Surround	-infinity to -1.5 dB	-3 dB	Default mix level for surround signals into L/R channels of Lt/Rt downmix
Lo/Ro Center	-6 to +3 dB	-3 dB	Default mix level for center signals into L/R channels of Lo/Ro downmix
Lo/Ro Surround	-infinity to -1.5 dB	-3 dB	Default mix level for surround signals into L/R channels of Lo/Ro downmix
Preferred stereo downmix	Pro Logic II	Default for Online Media	Instructs the decoder to perform a Lt/Rt downmix compatible with Dolby Pro Logic II, and Dolby Pro Logic IIx
	Surround	Default for Blu-ray Disc	Instructs the decoder to perform a Pro Logic Lt/Rt downmix
	Stereo		
	Not Indicated		

Related information

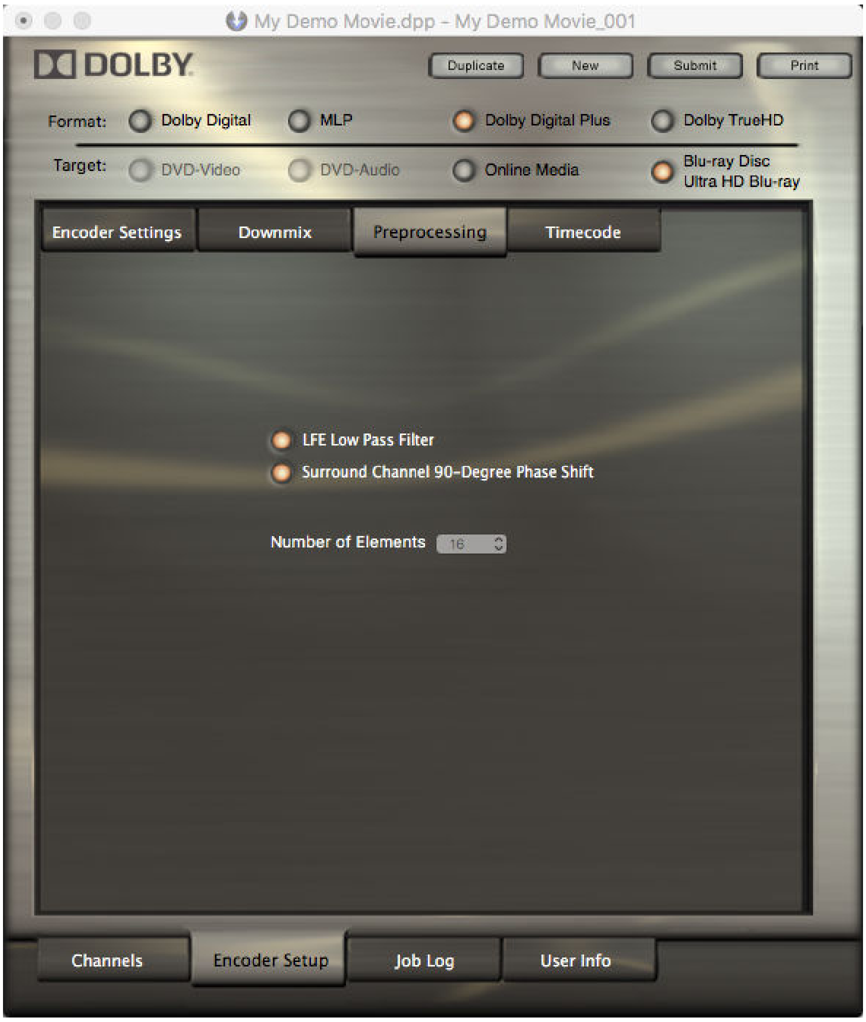
Downmixing in Dolby systems on page 83

6.4 Dolby Digital Plus preprocessing settings

On the **Preprocessing** tab under the **job editor Encoder Setup** tab you can modify the default values for several optional processes that can be applied to source files as part of the encoding process.

Figure 16: Dolby Digital Plus ***Preprocessing*** *tab*

*Figure 17: Dolby Digital Plus **Preprocessing** tab for Dolby Atmos encoding jobs*



LFE Low Pass Filter

The **LFE Low Pass Filter** option is available for Blu-ray Disc and Ultra HD Blu-ray media targets when the encoding configuration includes an LFE channel.

Enabling this setting applies a 120 Hz lowpass filter to the LFE input channel. This filter should always be enabled unless you are absolutely sure that your LFE source has no audio above 120 Hz.

The LFE lowpass filter is an eighth-order lowpass filter with unity gain at low frequencies and a steep rolloff at the upper edge of the LFE bandwidth (-3 dB above 120 Hz).

Surround Channel 90-Degree Phase Shift

The **Surround Channel 90-Degree Phase Shift** setting is only available when surround channels are present in the source.

The phase shift reduces undesirable signal cancellation in stereo presentations. The phase shift also makes it easier for many upmixers to extract the rear-channel information, resulting in better separation between the front and rear channels. It is especially useful if there are similar sounds in the L, R, Ls, and Rs source channels.

Surround Channel 3 dB Attenuation

This setting is available only when surround channels are present in the source and the source is a set of PCM channel signals.

The setting applies a 3 dB attenuation to the surround channel source files prior to encoding.

Note: Surround channel attenuation is appropriate when encoding a multichannel soundtrack created in a room with film-style calibration for consumer home theater playback. Channel-based cinema soundtrack surround channels are traditionally recorded 3 dB louder relative to the front channels, which requires that the surround loudspeakers are set 3 dB lower than the front (screen) channels. Home theater surround speakers are calibrated differently, with all speakers playing back at the same level.

Number of elements

This setting displays the number of clustered signals plus bed signals of Dolby Atmos material to be output by spatial coding when encoding rendered 7.1 or 5.1 audio from a Dolby Atmos master file set to channel-based Dolby Digital Plus bitstreams.
The number of elements depends on the selected Dolby Digital Plus media target:

- For **Online Media** with 384 kbps data rate, it is hardcoded to **12**.
- For **Online Media** with 448 kbps or higher data rates, it is hardcoded to **16**.
- For **Blu-ray Disc and Ultra HD Blu-ray**, it is always hardcoded to **16**.

6.5 Dolby Digital Plus timecode settings

The **Timecode** tab under the job editor **Encoder Setup** tab provides the same display and options for Dolby TrueHD, Dolby Digital Plus, and Dolby Digital encoding jobs.

Note: Timecode output is disabled when encoding Dolby Digital Plus streams for Online Media.

Related information

Dolby TrueHD timecode settings on page 46

7 MLP encoder setup options

The selected format and targets of an encoding job determine its available encoding configurations and settings.

- MLP encoding configurations
- MLP encoder settings
- MLP downmix settings
- MLP preprocessing settings
- MLP timecode settings

7.1 MLP encoding configurations

Dolby Media Encoder presents several encoding configurations for encoding MLP bitstreams.

Figure 18: MLP ***Channels*** *tab*

Table 8: MLP encoding configurations

Encoding configuration	Comments
1.0 - Mono	
2.0 - Stereo	
5.1 - L, R, C, LFE, Ls, Rs	Default configuration
3.0 - L, C, R	
3.1 - L, C, R, LFE	
4.0 - L, R, Ls, Rs	
5.0 - L, R, C, Ls, Rs	

Related information

Selecting encoding configuration and loading source files on page 21

7.2 MLP encoder settings

On the **Encoder Settings** tab under the **job editor Encoder Setup** tab you can modify default metadata parameters and other encoding settings.

Figure 19: MLP Encoder Settings tab

Use the **Restart Point in Access Units** setting to specify the number of access units between restart points. The default spacing for MLP Lossless audio is the maximum spacing, which uses one restart point every 32 access units, equivalent to a time interval of 27 ms.

Selecting a shorter interval between restart points allows decoding from midstream (after a pause, fast-forward, or rewind operation) to start in less than 27 ms, but the data rate of the MLP Lossless stream increases. An interval of 32 access units is appropriate for most applications.

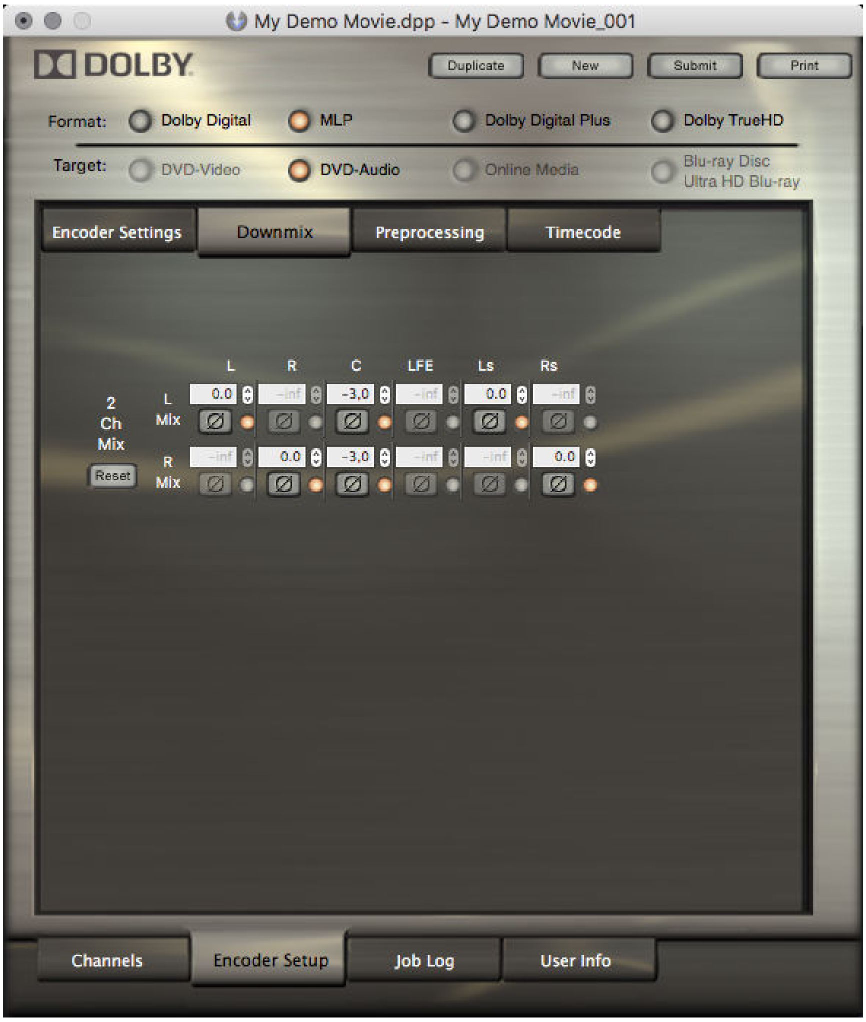
Related information

Configuring additional encoder settings on page 23

7.3 MLP downmix settings

On the **Downmix** tab under the **job editor Encoder Setup** tab you can select the channels to be included in the two-channel downmix and specify a separate attenuation for each channel.

Figure 20: MLP Downmix tab



Each channel includes a **Phase**  button to invert the phase of the signal.

Related information

Downmixing in Dolby systems on page 83

7.4 MLP preprocessing settings

On the **Preprocessing** tab under the **job editor Encoder Setup** tab, you can modify the default values for several optional processes that can be applied to source files as part of the encoding process.

Figure 21: MLP Preprocessing tab

Surround channel 3 dB attenuation
: When enabled, applies a 3 dB attenuation to the surround channel source files before encoding.

Apply apodization filter
: When applied, the apodization filter assists in minimizing the ringing effect that can occur in some analog-to-digital and digital-to-analog filters. Apodization filters can aid in decoding efficiency and improve the response of the audio through many A/D converters. The apodization filter is applicable only to source files with a sampling rate of at least 88.2 kHz.

7.5 MLP timecode settings

The **Timecode** tab under the **job editor Encoder Setup** tab provides options to specify a particular time range to encode.

Figure 22: MLP Timecode tab

Table 9: MLP encoding timecode options

Option	Values
Specify Time Range to Encode	On / Off
Encode Start	Valid timestamp in ms or frames
Encode End	Valid timestamp in ms or frames
Timecode Frame Rate	23.976 (fps)
	24 (fps)
	29.97 (fps)
	29.97 df (drop-frame fps)
	30 (fps)
	Real Time (ms)
	NTSC -0.1% (ms)
Prepend Silence	Valid number of frames or milliseconds *
Append Silence	Valid number of frames or milliseconds *

* The unit depends on the value of the **Timecode Frame Rate** field.

8 Dolby Digital encoder setup options

The selected format and targets of an encoding job determine its available encoding configurations and settings.

- Dolby Digital encoding configurations
- Dolby Digital encoder settings
- Dolby Digital downmix settings
- Dolby Digital preprocessing settings
- Dolby Digital timecode settings

8.1 Dolby Digital encoding configurations

Dolby Media Encoder presents several encoding configurations for encoding Dolby Digital bitstreams.

*Figure 23: Dolby Digital **Channels** tab*

Table 10: Dolby Digital encoding configurations

Encoding configuration	Comments
1.0 - Mono	
2.0 - Stereo	
2.0 - Lt/Rt	Flagged as matrix-encoded Lt/Rt signal. Dolby Media Encoder only sets metadata in the encoded stream so that a decoder can identify how the tracks should be played back.
5.1 - L, R, C, LFE, Ls, Rs	
5.1 Ex - L, R, C, LFE, Ls, Rs	Flagged as including a Dolby Surround EX content. Select this option for source files with additional matrix-encoded channels already mixed into the Ls and Rs inputs.
5.1 - L, R, C, LFE, Ls, Rs (Lrs, Rrs)	Default configuration. Lrs/Rrs source files are downmixed into the Ls/Rs channels before encoding, enabling additional downmix options. Select this option to produceDolby Digital 5.1-channel files from 7.1 source files.
3.0 - L, C, R	
3.1 - L, C, R, LFE	
4.0 - L, C, R, S	
4.0 - L, R, Ls, Rs	
4.1 - L, R, Ls, Rs, LFE	
5.0 - L, R, C, Ls, Rs	
Dolby Atmos render to 5.1 (Ls, Rs)	Encodes from a Dolby Atmos master file set to a 5.1 Dolby Digital bitstream without Dolby Atmos, enabling additional downmix options. Select this option to create a 5.1-channel Dolby Digital companion track for Blu-ray Disc.

Some Dolby Digital encoding configurations enable additional downmix options within the **Channels** tab.

Table 11: Dolby Digital additional downmixing options

Downmix option	Comments
5.1 Dolby PLIIx	Default option
5.1 Standard (Auto)	

The coefficients for a Dolby Pro Logic IIx (PLIIx) downmix are:
Ls = Lss + (-1.2 dB x Lrs) + (-6.2 dB x Rrs)
Rs = Rss + (-6.2 dB x Lrs) + (-1.2 dB x Rrs)

The coefficients for a Standard (Auto) downmix are:
Ls = 0 dB x Lss + 0 dB x Lrs
Rs = 0 dB x Rss + 0 dB x Rrs

Related information

Selecting encoding configuration and loading source files on page 21
Downmixing in Dolby systems on page 83

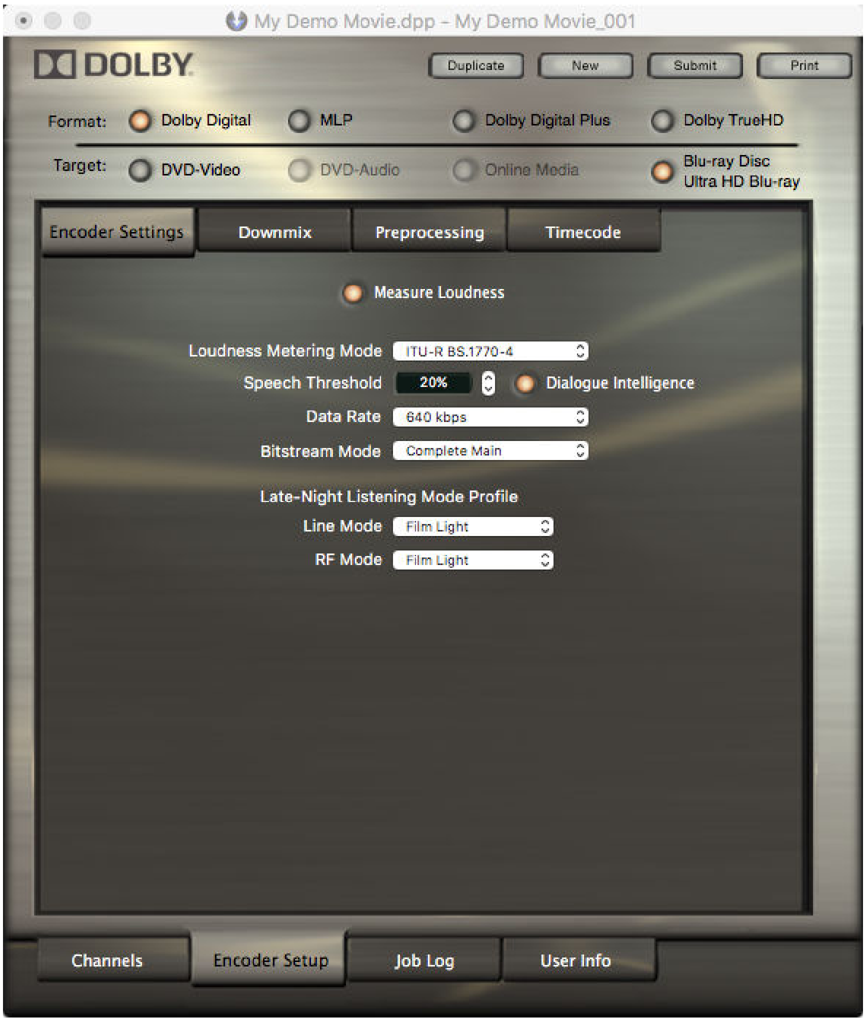
8.2 Dolby Digital encoder settings

On the **Encoder Settings** tab under the **job editor Encoder Setup** tab, you can modify default metadata parameters and other encoding settings, including Loudness measurement.

The **Encoder Settings** tab presents different options depending on whether the **Measure Loudness** option is enabled or disabled.

*Figure 24: Dolby Digital **Encoder Settings** tab with Loudness measurement disabled*

*Figure 25: Dolby Digital **Encoder Settings** tab with Loudness measurement enabled*



Loudness measurement options

Setting	Range
Loudness Metering Mode	ITU-R BS. 1770-4
	ITU-R BS. 1770-3
	ITU-R BS. 1770-2
	ITU-R BS. 1770-1
	Leq(A)
Dialogue Intelligence	Enabled (default)
	Disabled
Speech Threshold *	0% to 100% (default 20%)

* This setting is only active when **Dialogue Intelligence** is enabled.

- The **Loudness Metering Mode** setting specifies the weighting (A-weighting or K-weighting) used to estimate perceived loudness and whether dialogue should serve as the anchor element for measuring loudness. These modes include ITU standards and equivalent continuous A-weighted sound pressure level.
- The **Dialogue Intelligence** parameter enables or disables a feature that identifies and analyzes dialogue segments as a basis for speech gating and loudness measurement.

- The **Speech Threshold** setting defines the minimum amount of speech a program must contain in order for dialogue to be used as the anchor element for loudness measurements. When Dialogue Intelligence detects that the Speech Threshold is exceeded, the Loudness Metering Mode reverts to ITU-R BS. 1770-1 for the content identified as speech, that is, the silence gate and relative gate used in ITU-R BS. 1770-2, -3, and -4 are not used.

Note: These options are only available if the **Measure Loudness** option is enabled.

Other **Encoder Settings** options

Setting	Range
Dialogue level *	-1 to -31 (default -31)
Data rate	Based on the selected format, target, and number of input channels, Dolby Media Encoder automatically selects the most common data rate and provides a list of alternatives.
Bitstream Mode	Complete Main (default)
	Music & Effects
	Visually Impaired
	Hearing Impaired
	Dialogue
	Commentary
	Emergency
	Voice Over
Late-Night Listening Mode Profile, Line Mode	Film Light (default)
	Film Standard
	Music Standard
	Music Light
	Speech
Late-Night Listening Mode Profile, RF Mode	Film Light (default)
	Film Standard
	Music Standard
	Music Light
	Speech

* Not available if the **Measure Loudness** option is enabled.

Related information

Configuring additional encoder settings on page 23

8.3 Dolby Digital downmix settings

On the **Downmix** tab under the **job editor Encoder Setup** tab you can modify the default metadata used for instructing a decoder on how to downmix its output.

*Figure 26: Dolby Digital **Downmix** tab*

The options defined on the **Downmix** tab are different from the downmix options on the **Channels** tab and are meant to instruct decoders on how to downmix for two-channel outputs. The Lx/Rx mix-level parameters let the content creator set the levels at which the Center and Surround signals are mixed into the L and R outputs when downmixing is active in a decoder product. The levels can be set independently for the surround-encoded (Lt/Rt) and stereo (Lo/Ro) downmix types.

Setting	Range	Default value	Comments
Lt/Rt Center	-6 to +3 dB	-3 dB	Default mix level for center signals into L/R channels of Lt/Rt downmix
Lt/Rt Surround	-infinity to -1.5 dB	-3 dB	Default mix level for surround signals into L/R channels of Lt/Rt downmix
Lo/Ro Center	-6 to +3 dB	-3 dB	Default mix level for center signals into L/R channels of Lo/Ro downmix
Lo/Ro Surround	-infinity to -1.5 dB	-3 dB	Default mix level for surround signals into L/R channels of Lo/Ro downmix

Setting	Range	Default value	Comments
Preferred stereo downmix	Surround	Default	Instructs decoders set to the Auto downmix selection to enable a particular downmix
	Stereo		
	Not Indicated		

Related information

Downmixing in Dolby systems on page 83

8.4 Dolby Digital preprocessing settings

On the **Preprocessing** tab under the **job editor Encoder Setup** tab you can modify the default values for several optional processes that can be applied to source files as part of the encoding process.

*Figure 27: Dolby Digital **Preprocessing** tab*

LFE Lowpass Filter

This option is available when the encoding configuration includes an LFE channel.

Enabling this setting applies a 120 Hz lowpass filter to the LFE input channel. This filter should always be enabled unless you are absolutely sure that your LFE source has no audio above 120 Hz.

The LFE lowpass filter is an eighth-order lowpass filter with unity gain at low frequencies and a steep rolloff at the upper edge of the LFE bandwidth (-3 dB above 120 Hz).

Surround Channel 90-Degree Phase-Shift

This setting is only available when surround channels are present in the source.

The phase shift reduces undesirable signal cancellation in stereo presentations. The phase shift also makes it easier for many upmixers to extract the rear-channel information, resulting in better separation between the front and rear channels. It is especially useful if there are similar sounds in the L, R, Ls, and Rs source channels.

Surround Channel 3 dB Attenuation

This setting is available only when surround channels are present in the source and the source is a set of PCM channel signals.

The setting applies a 3 dB attenuation to the surround channel source files before encoding.

Note: Surround channel attenuation is appropriate when encoding a multichannel soundtrack created in a room with film-style calibration for consumer home theater playback. Channel-based cinema soundtrack surround channels are traditionally recorded 3 dB louder relative to the front channels, which requires that the surround loudspeakers are set 3 dB lower than the front (screen) channels. Home theater surround speakers are calibrated differently, with all speakers playing back at the same level.

8.5 Dolby Digital timecode settings

The **Timecode** tab under the **job editor Encoder Setup** tab provides the same display and options for Dolby TrueHD, Dolby Digital Plus, and Dolby Digital encoding jobs.

Related information

Dolby TrueHD timecode settings on page 46

9 Management of encoding jobs and projects

Dolby Media Encoder handles all the information used to create an encoded file within an encoding job. Encoding jobs files are created and managed within encoding projects.

9.1 Project usage

A project can help you manage your encoding jobs and make it easy to find previously encoded job edits. This includes letting you review job logs, encoder settings, and files used.

Here are some common ways that you can use a project:

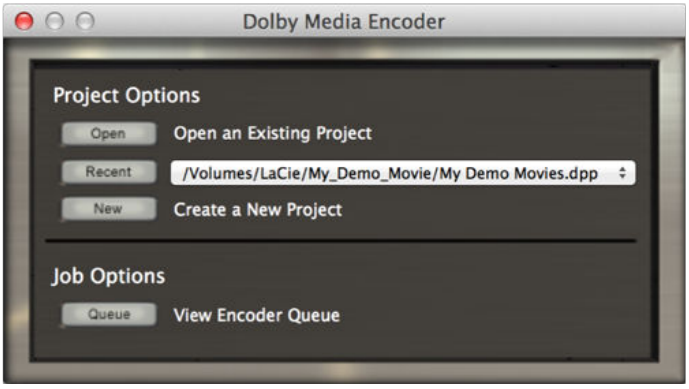
- Create a project for a particular movie, including all the encoding jobs for that movie within that project. All of the job encoding information for that movie is kept together.
- Create a project file for a particular customer, including all the encoding jobs for that customer within that project. All of the job encoding information for that customer is kept together.
- Create a project for each day or week in your facility, including all the encoding jobs for that day or week within that project. All of the job encoding information for that day or week is kept together.
- Create a separate project for each user, including all the encoding jobs submitted by each user within their respective projects. All of the job encoding information for each user is kept separate.
- Create a separate project for each encoding job you do so that the project file contains all of the information for that one job.

9.2 Creating a new project

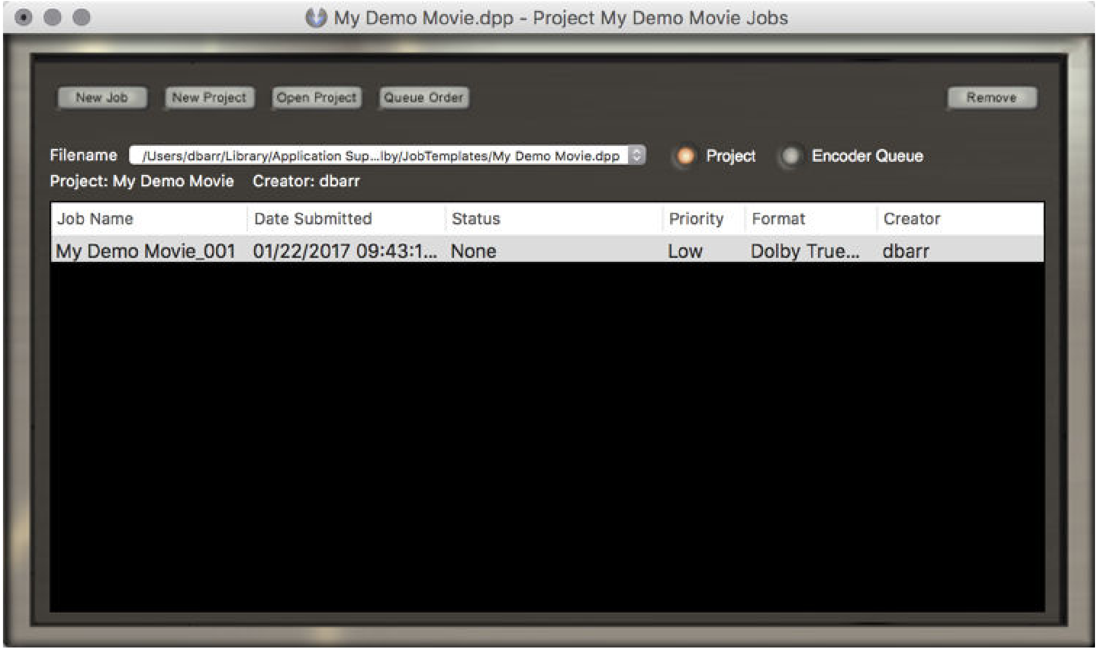
When you create a new project and save it, you automatically create a new job that you can configure for encoding.

Procedure

1. Access the **Project properties** window.
    - You can create a new project from the Dolby Media Encoder **initial window** by clicking **New** in the **Project Options** section.



- You can create a new project from an existing project in the job list window by clicking **New Project** at the top of the window.



The system displays the **Project properties** window.

2. In the **Project Name** field, enter a name for the project.
3. Enter additional information fields.

   Facility (optional)
   This field lets you designate the site where the encoding work was done. After creating a second project, the field is automatically filled with the last value you entered.

   Select Job Template
   You can browse to a .djt template file or enter its location here. Dolby Media Encoder will use that template for each new job you create in that project. Templates are always stored in a system template location.

   Clicking the  (browse) button always takes you to that location automatically. Doing this makes templates more user-friendly for future use.

   Comments
   This field provides an area to enter information for later reference.

4. Click **Create**.
   The system displays a **Save** dialog box.
5. Save the project .dpp file at any location of your choice.

   The file name can be different from the project name.
6. Click **OK**.
   The system opens the window for the new project.

What to do next
You can view and change the properties of any projects by selecting **File** > **Project Properties** on the Dolby Media Encoder Client menu bar.

Related information

Creating job copies for reuse on page 78

9.3 Managing jobs in project view

After a project has been created and saved, you can manage its jobs from the job list window.

- You can use the job list window to access existing projects.
  - On the menu bar, select **File** > **Open Project**, browse to an existing project, select it, and then click **Open** in the dialog box.
  - On the menu bar, select **File** > **Open Recent** and select a recent project.
  - Click **Open Project**, browse to an existing project, select it, and then click **Open** in the dialog box.
- You can use the job list window to sort existing projects.
  a) Click the heading of any column to sort the list by that field.
  b) Click **Queue Order** at the top of the window to restore the initial sorting order.
- You can use the **job list** window to open or remove any existing jobs within the current project.
  - Right-click a job and select **Open** or **Remove**.
  - From the list, select one of more jobs, and then click **Remove** at the top of the job list window.

    Note: The encoder supports selecting one job by clicking (highlighting) it or selecting multiple jobs by clicking (highlighting) a job and then selecting additional jobs by using the `<Shift>` or `<Command>` key.

9.4 Managing jobs in the encoder queue

You can view and manage submitted jobs in the encoder queue of the job list window.

Prerequisites
In the job list, click **Encoder Queue**.

About this task

You can see all the jobs that are currently encoding, queued, or completed by the selected server. You can pause jobs, move them up or down in the queue, and remove them. You can also open a submitted job and duplicate it for use in a new encoding job.

- You can filter encoding jobs by their status.
  - Click **Active** to view active jobs.
  - Click **Archived** to view encoding jobs that have been completed and auto-archived.
- You can change the order of jobs in the job list.
  - Click the heading of any column to sort by that value.
  - Click **Queue Order** to return the page to the default sort order.
  - Move jobs up and down in the list using the **Move Up** and **Move Down** buttons.
- You can move, pause, resume, or remove jobs in the encoder queue by using action buttons or right-click commands.

Only jobs that are not yet encoded can be moved, paused, or resumed.

Note: The encoder supports selecting one job by clicking (highlighting) it, or selecting multiple jobs by clicking (highlighting) a job and then selecting additional jobs by using the `<Shift>` or `<Command>` key.

- Click one of the encoder queue action buttons.

  Move Up
  Moves the highlighted jobs up one position in the queue.

  Move down
  Moves the highlighted jobs down one position in the queue.

  Pause / Resume
  Pauses and resumes processing of the highlighted jobs.

  Remove
  Removes the highlighted jobs from the queues.

- Right-click jobs in the encoder queue.

  Move Up
  Moves the highlighted jobs up one position in the queue.

  Move down
  Moves the highlighted jobs down one position in the queue.

  Move top
  Moves the highlighted job to the top of its priority group.

  Move bottom
  Moves the highlighted job to the bottom of its priority group.

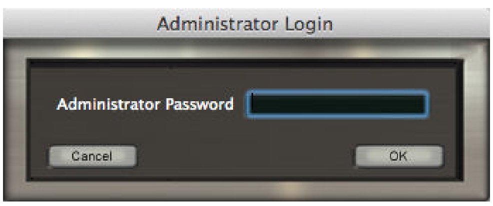
  Pause / Resume
  Pauses and resumes processing of the highlighted jobs.

  Open
  Opens the job editor window for the highlighted job.

  Remove
  Removes the highlighted jobs from the queues.

- You can manage jobs submitted on the local client or submitted by someone else on a different client.

  If you attempt to change a job submitted by another user, the system requires you to enter the administrative password before it makes the change. The default password is `Dolby`.



Note: The **Administrator Password** field is case sensitive.

9.5 Encoding files through Hot Job folders

A Hot Job is a quick access folder that acts like an application that automates your encoding workflow. You can drag and drop sets of source files into a Hot Job folder to initiate encoding jobs.

About this task

The Hot Job automatically sets up and queues jobs using the chosen settings. Hot Jobs can be used to automatically set up as many separate encoding jobs as you like.

Note: Hot jobs are not supported in remote encoding jobs. Dolby Media Encoder Client must be set up to use the local version of Dolby Media Encoder Server.

Procedure

1. On the application menu bar, select **Job** > **New Hot Job**.

   If you already have an open Hot Job, select **Job** > **Duplicate**.

   A **job editor** window opens.
2. If needed, edit the Hot Job folder name in the **Job Name** field in the **User Info** page of the **job editor**.

   The default name is HotJob.app, but you can change it. Source file section is unavailable because the source files you use will be the ones dropped into the folder.
3. Select a destination path.

   This is the location for your encoded jobs.
4. On the Encoder Setup pages, select configuration settings that suit your workflow.

   Select those settings that apply to all jobs generated by the Hot Job.
5. Click **Save**.
   The new Hot Job folder appears on the desktop.

Related information

Encoding workflow in Dolby Media Encoder on page 18
Remote Dolby Media Encoder Server setup on page 11

9.6 Creating job copies for reuse

With the duplication and template features of Dolby Media Encoder, you can make copies of the settings for a job on the fly and to save them separately for reuse.

- Duplicate a job (or Hot Job) to reuse its configuration settings.

  Tip: Making a duplicate is the only way to modify the settings of a completed encoding job.

  a) Open the job (or Hot Job) you wish to duplicate in a job editor window.

  b) At the top of the job editor window, click **Duplicate**.

     Alternatively, select **Job** > **Duplicate Job** on the menu bar.

  The system opens a job with identical settings and _copy appended to its name. All configuration settings are the same, but the destination path is empty.
- Save an encoding job as a job template.

  Saving common job settings as a template saves you time by eliminating the need to manually set up all of the configuration options.

a) Open the job (completed or new) you wish to save as a template.

b) Select **Job** > **Save as Template**.

The system opens a standard **Save** dialog box. If you have files loaded into the job, their file names and paths will be included in the saved template.

c) Enter a name for the template, and select a destination for it.

The **Save** dialog box remembers the previous destination location. We recommend storing templates under //Users/<username>/Library/Application Support/Dolby/JobTemplates/.

d) Click **Save**.
The system saves the file with a .djt extension.

- Optional: Apply job templates to an entire project by loading the template file into the **Select Job Template** field in the **Project Properties** window.

  When a template is selected, new jobs opened in that project will use that template instead of the application default.

- Optional: Apply a job template for an individual job by selecting **Job** > **Load Template** on the application menu bar.

  Selecting **Load Template** lets you select a saved .djt file from the folder of your choice. Once you load a template, a new job editor opens with the settings for the template.

- Optional: You can delete job templates by selecting **Job** > **Delete Template**.

  Selecting **Delete Template** lets you select and delete a saved .djt file from the folder of your choice.

Related information

Creating a new project on page 73

9.7 Managing the encoder log

Dolby Media Encoder automatically creates a log for each job. The log lists all successful step completions and any error conditions that occur during encoding.

Procedure

To access the encoder log, click the **Job Log** tab of the job editor window.

9.8 File-name conventions for automatic file selection

Dolby Media Encoder can automatically load source file sets that follow a particular file-name convention.

- Uses the following channel IDs:

Label	Channel
L, Lt	Left
R, Rt	Right
C	Center
LFE, LF, SW	Low-frequency effects
Ls, Ssl, Lss	Left surround
Rs, Ssr, Rss	Right surround

Label	Channel
Lc	Left center
Rc	Right center
Lrs, Rls, Bsl, Lsr	Right rear surround
Cs	Center surround
S	Surround
Lsd	Left Surround direct
Rsd	Right surround direct
Lw	Left wide
Rw	Right wide
Lvh, Vhl, Lh	Vertical height left
Vhc, Ch	Vertical height center
Rvh, Vhr, Rh	Vertical height right

- Channel IDs are not case sensitive.
- The channel ID is separated before and after with either a period or an underscore.
- The file names for the channels are identical except for channel ID.
- The number of files matches the number of channels selected on the **Channels** tab of the job editor window.
- The file channel IDs match the channel IDs selected on the **Channels** tab of the job editor window.
- File names do not contain the ampersand (&) character.

Examples

Using a period separator:

```
ExampleOne.L.wav
ExampleOne.R.wav
ExampleOne.C.wav
ExampleOne.LFE.wav
ExampleOne.Ls.wav
ExampleOne.Rs.wav
```

Using a period separator, with extra information after the channel ID:

```
ExampleOne.L.ENG.wav
ExampleOne.R.ENG.wav
ExampleOne.C.ENG.wav
ExampleOne.LFE.ENG.wav
ExampleOne.Ls.ENG.wav
ExampleOne.Rs.ENG.wav
```

Using an underscore separator:

```
ExampleTwo_L.wav
ExampleTwo_R.wav
ExampleTwo_C.wav
ExampleTwo_LFE.wav
```

ExampleTwo_Ls.wav
ExampleTwo_Rs.wav

Using an underscore separator, with extra information after the channel ID:

ExampleTwo_L_ENG.wav
ExampleTwo_R_ENG.wav
ExampleTwo_C_ENG.wav
ExampleTwo_LFE_ENG.wav
ExampleTwo_Ls_ENG.wav
ExampleTwo_Rs_ENG.wav

Related information

Selecting encoding configuration and loading source files on page 21

10 Reference information

Dolby Media Encoder leverages particular Dolby technologies.

- Dynamic range control
- Downmixing in Dolby systems
- Dolby Atmos technology
- Dolby TrueHD technology
- Dolby Digital Plus technology
- MLP Lossless technology
- Dolby Digital technology
- Dolby Surround / Dolby Pro Logic technology
- Dolby Pro Logic II technology
- Dolby Surround EX technology
- Dolby Pro Logic IIx technology
- Dolby headphones technology
- Loudness measurement

10.1 Dynamic range control

Dynamic range control applies to Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstreams during decoding.

Decoders interpret dynamic range control metadata to control compression for two-channel downmix overload protection mode and late-night listening modes.

Encoders include dynamic range control metadata in the bitstream to direct the behavior of the compressor in the decoder. The Dolby Media Encoder includes the following dynamic range control presets:

Film Light (default)

Provides moderate compression parameters in the bitstream that the listener may choose to apply during playback.

- Max boost: 6 dB (below –53 dB)
- Boost range: –53 to –41 dB (2:1 ratio)
- Null band width: 20 dB (–41 to –21 dB)
- Early cut range: –21 to –11 dB (2:1 ratio)
- Cut range: –11 to +4 dB (20:1 ratio)

Film Standard

- Max boost: 6 dB (below –43 dB)
- Boost range: –43 to –31 dB (2:1 ratio)
- Null band width: 5 dB (–31 to –26 dB)
- Early cut range: –26 to –16 dB (2:1 ratio)
- Cut range: –16 to +4 dB (20:1 ratio)

Music Standard

- Max boost: 12 dB (below –55 dB)
- Boost range: 55 to –31 dB (2:1 ratio)
- Null band width: 5 dB (–31 to –26 dB)
- Early cut range: –26 to –16 dB (2:1 ratio)
- Cut range: –16 to +4 dB (20:1 ratio)

Music Light

- Max boost: 12 dB (below –65 dB)
- Boost range: –65 to –41 dB (2:1 ratio)
- Null band width: 20 dB (–41 to –21 dB)
- Early cut range: N/A
- Cut range: –21 to +9 dB (2:1 ratio)

Speech

- Max boost: 15 dB (below –50 dB)
- Boost range: –50 to –31 dB (5:1 ratio)
- Null band width: 5 dB (–31 to –26 dB)
- Early cut range: –26 to –16 dB (2:1 ratio)
- Cut range: –16 to +4 dB (20:1 ratio)

Compression choices are available for both Line mode and RF mode. The content producer chooses which of these profiles to assign to each mode. These profiles are applied when the decoder selects a dynamic range control mode.

Dynamic range control profile settings are dependent on an accurate dialogue level setting. Improper setting of the dialogue level parameter may result in excessive and audible application of overload-protection limiting.

The dialogue level value determines the bitstream `dialnorm` parameter, and sets the position of the compressor null band where the signal is neither boosted nor attenuated. The null band is designed to protect audio at the average level of the program from having either cut or boost applied when dynamic range control is applied in the decoder.

Dynamic range control metadata provides graceful control of excessive dynamic range and overload protection. For example, consider a 5.1-channel program with signals at digital full scale on all channels being played through a stereo, downmixed line-level output: without some form of attenuation or limiting, such output signal would certainly clip. Accurately setting the dialogue level and dynamic range control profiles prevents clipping and the unnecessary application of automatic overload protection.

10.2 Downmixing in Dolby systems

Downmixing settings in Dolby Media Producer Master Suite are based on the encoding format, target, and expected listening environments.

Stereo downmixing

The following stereo downmix options are supported:

Dolby TrueHD
Standard: Two-channel stereo Left, Right (Lo/Ro)
Custom: Includes adjustable controls for setting levels

Dolby Digital Plus and Dolby Digital
Surround: Two-channel Left, Right (matrix-encoded Lt/Rt signal)
Stereo: Two-channel stereo Left, Right (Lo/Ro)
Dolby Pro Logic II: Two-channel Left, Right (Dolby Pro Logic II matrix-encoded Lt/Rt signal)

Dolby Digital Plus and Dolby Digital encoding can apply phase 90 filters. The phase 90 filters used by default provide the all-pass 90-degree phase shift filtering for the Ls/Rs feeds into the downmix. This reduces undesirable signal cancellation in stereo presentations, improves imaging, and enables proper matrix decoding. It is strongly recommended to use the 90-degree phase shift for any Lt/Rt downmixes.

Stereo downmixing involves an initial 7.1 to 5.1 downmix, followed by a 5.1 to 2.0 downmix in the same manner as 2.0 consumer products. The coefficients for the two-channel downmixes from 5.1 are as follows:

Lt/Rt (legacy)
Lt = L + (–3 dB x C) – (–3 dB x Ls) – (–3 dB x Rs)
Rt = R + (–3 dB x C) + (–3 dB x Ls) + (–3 dB x Rs)

Lt/Rt (Pro Logic II)
Lt = L + (–3 dB x C) – (–1.2 dB x Ls) – (–6.2 dB x Rs)
Rt = R + (–3 dB x C) + (–6.2 dB x Ls) + (–1.2 dB x Rs)

LoRo
Lo = L + (–3 dB x C) + (–3 dB x Ls)
Ro = R + (–3 dB x C) + (–3 dB x Rs)

10.3 Dolby Atmos technology

The Dolby Atmos experience combines a channel-based audio bed with object-oriented sound to place and move specific effects around the listener.

Using dynamic audio objects, sound designers and artists are free to mix audio in a three-dimensional space, steering effects throughout the room. When a Dolby Atmos soundtrack is played back in the cinema, the object-based audio mix is adapted to the room: audio is rendered to the exact speaker layout of the theatre.

A major benefit of Dolby Atmos is the ability to scale a soundtrack to different room sizes and speaker configurations. Dolby Atmos provides an optimal presentation of the soundtrack for each listening environment.

Audio/video receiver and home-theater-in-a-box products were the first consumer products to support Dolby Atmos. These products decode, scale, and calibrate Dolby Atmos content to accommodate a variety of home theater configurations.

Dolby Atmos for home theater has the following features to produce a three-dimensional audio experience:

- Object-based Dolby Atmos content
- Dolby Atmos decoders
- Dolby advanced audio postprocessing
- Dolby Atmos enabled speaker technology

10.4 Dolby TrueHD technology

Dolby TrueHD is a lossless coding technology for high-definition disc-based media.

Dolby TrueHD delivers sound that is identical to the studio master, unlocking the true high-definition entertainment experience on current and future surround formats on Blu-ray Disc.

Features

- Lossless coding technology.
- Supports carriage of Dolby Atmos content.
- Supports data rates up to 18 Mbps.
- Supports up to eight full-range channels of 24-bit/96 kHz audio and up to six channels of 24-bit/192 kHz audio.
- Supported by HDMI.
- Supports extensive metadata, including dialogue normalization and dynamic range control.
- Supports multiple discrete presentations that can be custom mixes for Dolby Atmos, 7.1, 5.1, or stereo.

Benefits

- Studio master-quality sound that unlocks the true high-definition entertainment experience on next-generation discs.
- Compatible with Dolby Digital audio/video receivers and home-theater-in-a-box products.
- Dialogue normalization that can maintain the same volume level when you change to other Dolby TrueHD , Dolby Digital Plus, and Dolby Digital programming.
- Dynamic range control features enable the customization of audio playback to reduce peak volume levels while experiencing all the details in the soundtrack.
- Selected as a format for Blu-ray Disc.

10.5 Dolby Digital Plus technology

Dolby Digital Plus combines the efficiency to meet future broadcast demands with the power and the flexibility to realize the full audio potential of the high-definition experience.

Built over the foundation of Dolby Digital, Dolby Digital Plus provides even higher quality audio, more channels, and greater flexibility, but remains fully compatible with all current audio/video receivers.

Features

- Multichannel sound with discrete channel output.
- Supports carriage of Dolby Atmos.
- Channel and program extensions can carry multichannel audio programs of up to 7.1 channels and support multiple programs in a single encoded bitstream.
- Outputs a Dolby Digital bitstream for playback on existing Dolby Digital systems.
- Supports data rates as high as 6 Mbps.
- Bit-rate performance of up to 1.6 Mbps on Blu-ray Disc.

- Interactive mixing and streaming capability in advanced systems.
- Supported by HDMI.

Benefits

- Can deliver 7.1 channels or more enhanced-quality audio at up to 6 Mbps.
- Allows multiple languages to be carried in a single bitstream.
- Offers audio professionals new creative power and freedom.
- Compatible with the millions of home entertainment Dolby Digital systems.
- No latency or loss of quality in the conversion process. Maintains high quality at more efficient broadcast bit rates (<192 kbps for 5.1-channel audio).
- Selected by the Advanced Television Systems Committee (ATSC) as the standard for future broadcast applications; named as an option by the Digital Video Broadcasting (DVB) project for satellite and cable TV.
- Selected as a format for Blu-ray Disc.

10.6 MLP Lossless technology

MLP Lossless, licensed by Dolby Laboratories, is the core technology of Advanced Resolution multichannel and stereo DVD-Audio.

MLP Lossless enables producers to encode up to six channels of 24-bit/96 kHz audio or two channels of 24-bit/192 kHz audio onto a DVD-Audio disc, resulting in playback that is bit-for-bit identical to the studio master. Nothing is lost during the encoding and decoding process.

Widely recognized as the first major step forward in high-fidelity audio after the compact disc, DVD-Audio makes your home audio system sound like a live musical performance. DVD-Audio combines DVD technology with MLP Lossless coding to give you music with full 5.1-channel surround sound and fidelity, which is dramatically better than that of the CD. A DVD-Audio player provides an excellent listening experience for DVD-Audio discs, but many DVD-Video player can also play DVD-Audio discs.

10.7 Dolby Digital technology

In thousands of cinemas and millions of homes worldwide, Dolby Digital is the standard for surround sound technology in general and 5.1-channel surround sound in particular.

Dolby Digital is a highly sophisticated and versatile audio encoding/decoding technology. Dolby Digital technology can transmit mono, stereo (two-channel), or up to 5.1-channel surround sound (discrete multichannel audio).

The superior coding efficiency of Dolby Digital, and its ability to deliver high-quality discrete multichannel audio without compromising video quality, has made it the designated audio standard for DVD worldwide.

Dolby Digital is also the preferred multichannel audio standard for broadcast direct to home satellite and digital cable systems. Nearly 60 million digital cable and satellite set-top boxes are currently in use worldwide, supported by over 30 million audio/video receivers equipped with 5.1-channel Dolby Digital decoding. Virtually every DVD-Video player sold today offers a two-channel downmix of the Dolby Digital 5.1-channel movie soundtrack, as well as a digital output for connection to a 5.1-channel audio/video receiver.

In the film industry, the Dolby Digital soundtrack is optically encoded right on the filmstrip, in the space between the sprocket holes. Placing the soundtrack directly on the film allows it to coexist with the analog track, a cost-effective and simple solution for film distributors and theatre owners. The sprocket hole area has also proven highly resistant to wear and tear, so that a soundtrack encoded in Dolby Digital remains free of pops and hiss for the useful life of the print.

The sound information contained in each of the available Dolby Digital channels is distinct and independent. These six channels are described as a 5.1-channel system, because there are five full-bandwidth channels (Left, Right, Center, Left Surround, and Right Surround) with a 3 Hz to 20 kHz frequency range, plus a Low-Frequency Effects (LFE) subwoofer channel devoted to frequencies from 3 to 120 Hz.

10.8 Dolby Surround / Dolby Pro Logic technology

Dolby Surround is the consumer version of the original Dolby multichannel analog film sound formats (Dolby Stereo and Dolby SR).

When a Dolby Surround soundtrack is produced, four channels of audio information (Left, Center, Right, and mono Surround) are matrix encoded onto two audio tracks. These two tracks are then carried on stereo program sources such as stereo soundtracks on DVDs and Blu-ray Disc, videotapes, and TV broadcasts into the home, where they can be decoded by Dolby Pro Logic to re-create the original multichannel surround sound experience.

Dolby Pro Logic was the foundation for multichannel home theater, and was the reference decoder for creating the surround sound audio tracks in thousands of commercially available videocassettes, laser discs, DVDs, and television programs.

With the introduction of the Dolby Digital multichannel film sound format, Dolby Digital replaced Dolby Surround as the preferred technology to deliver multichannel audio to consumers via DVD-Video, digital television, and games. However, every Dolby Digital decoder also provides a stereo signal compatible with Dolby Pro Logic on its analog outputs.

10.9 Dolby Pro Logic II technology

Dolby Pro Logic II technology processes any high-quality stereo (two-channel) audio into five playback channels of full-bandwidth surround sound.

Dolby Pro Logic II is matrix surround decoding technology that detects the directional cues that occur naturally in stereo content, using these elements to create a five-channel surround sound playback experience. Dolby Pro Logic II is ideally suited for home theater systems, PCs, game consoles, and multichannel in-car audio systems.

Dolby Pro Logic II is fully compatible with all Dolby Pro Logic technologies. It provides optimal audio for playback in a 5.1-channel home theater system for the thousands of videocassettes and TV programs encoded in four-channel Dolby Surround (the encoding counterpart to the Dolby Pro Logic decoding technology).

Dolby Pro Logic II also enables video game consoles to encode five-channel surround sound information into a stereo signal with virtually no impact on the console CPU, which means better audio with no impact on game play.

In addition to enhancing playback, Dolby Pro Logic II can be used to encode TV programming to deliver a surround sound experience for viewers with stereo TV systems.

10.10 Dolby Surround EX technology

Dolby Surround EX augments the standard 5.1-channel cinema experience with a new Back Surround channel, reproduced by speakers placed directly behind the audience.

The Dolby Surround EX 6.1-channel format added a new dimension to the Dolby Digital 5.1 cinematic experience, and was soon introduced in DVD releases and 6.1 home theater playback equipment as well. The additional channel was matrix-encoded onto the regular Left Surround and Right Surround channels of a conventional Dolby Digital 5.1 soundtrack. This allowed it to be either matrix-decoded by a Dolby Surround EX system, or reproduced by the traditional Left Surround and Right Surround speakers in an older 5.1 system.

An EX flag in the bitstreams allows home theater equipment to detect content with this matrix-encoded Back Surround channel and reproduce it correctly.

The introduction of Dolby Pro Logic IIx 7.1 channel home theater systems surpassed the 6.1 experience and gained rapid adoption. The Dolby Pro Logic IIx system matrix encodes not one but two rear surround signals into the conventional Ls and Rs channels.

The EX flag in the bitstream is re-purposed to indicate presence of any matrix-encoded rear surround signals in a Dolby TrueHD, Dolby Digital Plus, or Dolby Digital 5.1-channel presentation, including the 5.1-channel presentation heard when decoding a 7.1-channel bitstream on a system with only 5.1 capability.

10.11 Dolby Pro Logic IIx technology

Dolby Pro Logic IIx was designed to expand upon several multichannel technologies, including Dolby Pro Logic II matrix decoding as well as Dolby Digital and Dolby Digital EX audio systems by deriving an additional pair of independent rear surround channels.

The additional Dolby Pro Logic IIx channel pair offers decoders additional flexibility in deriving up to 7.1 channels to feed all available speaker outputs of a surround sound system, and to do so from a wide range of sources (from stereo to discrete multichannel).

Whereas 7.1 rear surround channels add an extra dimension to surround sound systems, all Dolby Pro Logic IIx outputs are intended for speaker locations fixed on a horizontal plane.

Features

- Ability to generate up to 7.1 channels from stereo and 5.1 sources.
- Support of several decoding modes:
  - Music mode
  - Movie mode
  - Matrix mode
  - Dolby Digital EX mode
- Support for upmixed output configurations based on stereo sources:
  - 2 to 3 (L and R to L, C, and R)
  - 2 to 4 (L and R to L, R, Ls, and Rs)
  - 2 to 5.1 (L and R to L, C, R, Ls, and Rs)

  - 2 to 6.1 (L and R to L, C, R, Ls, Rs, and Cs)
  - 2 to 7.1 (L and R to L, C, R, Ls, Rs, Lrs, and Rrs)
- Support for upmixed output configurations based on 5.1 sources with Independent Ls and Rs channels:
  - 5.1 to 6.1 (Ls and Rs to Ls, Rs, and Cs)
  - 5.1 to 7.1 (Ls and Rs to Ls, Rs, Lrs, and Rrs)

Note: For 5.1-channel sources, Dolby Pro Logic IIx operates on the Ls and Rs channels only; L, C, R, and LFE channels are bypassed.

10.12 Dolby headphones technology

Dolby Headphone technology allows users to wear any set of headphones and listen to music, watch movies, or play video games with the dramatic surround effects of a 5.1-channel soundtrack.

Features

- Realistic 5.1-channel surround sound from any set of headphones.
- Processes 24-bit/96 kHz audio, such as DVD-Audio, to produce surround sound bit-for-bit identical to the studio masters.
- Superb surround sound with Dolby Pro Logic II from stereo music and movies.

Benefits

- Enables enjoyment of surround sound without disturbing others.
- Improves listening comfort and reduces listener fatigue.
- Allows stereo content to be heard as 5.1-channel surround sound (when combined with Dolby Pro Logic II).

10.13 Loudness measurement

Dolby Media Encoder can automatically measure loudness for Dolby TrueHD, Dolby Digital Plus, and Dolby Digital encoding jobs.

When loudness measurement is active, the encoder measures the loudness of the content, sets the `dialnorm` dialogue normalization value according to this measurement, and includes Intelligent Loudness metadata into the encoded bitstream.

When applying loudness measurement you can select between a selection of loudness metering mode, with reference to loudness metering ITU standards. The loudness metering mode determines the weighting and gating used to estimate perceived loudness.

Loudness metering modes
ITU-R BS. 1770-4
ITU-R BS. 1770-3
ITU-R BS. 1770-2
ITU-R BS. 1770-1
Leq(A) *

* Equivalent continuous A-weighted sound pressure level

You can also choose to use Dialogue Intelligence, a feature that identifies and analyzes dialogue segments within audio as a basis for speech gating, and you can specify the speech threshold, that is, the minimum amount of speech a program must contain in order for dialogue to be used as the anchor element for measuring loudness.

10.13.1 Intelligent loudness

Intelligent loudness steering facilitates loudness management within a broadcast chain. It addresses the problem of multiple stages of dynamic range processing within the chain that can lead to degradation of the audio quality.

Intelligent loudness metadata is carried alongside an audio signal in most Dolby bitstreams. It indicates to downstream devices whether the `dialnorm` value is known to be correct. In this way, devices that support intelligent loudness in the broadcast chain can identify audio that has been correctly leveled, and disable further leveling.

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- Disclaimers
- Open-source software

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The Dolby Media Encoder open-source software acknowledgments include the license and its corresponding component.

Table 12: Dolby Media Encoder open-source software

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Table 12: Dolby Media Encoder open-source software (continued)

License name	License URL	Component (module) name
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MIT License for BBC Audio Toolbox	https://www.bbc.co.uk/rd/publications/audio-definition-model-software	BBC Audio Toolbox
MIT License	https://github.com/jbeder/yaml-cpp/blob/master/LICENSE	Yaml-cpp v0.5.3
Mozilla Public License, Version 2.0	https://www.mozilla.org/en-US/MPL/2.0/	Eigen v3.1.3
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